

V810 S2 EX

Confirmation, Diagnostics & Adjustments (CD&A)





Confirmation, Diagnostics & Adjustments

1. **Confirmation** : Verify working subsystems
2. **Diagnostics** : Diagnose problem
3. **Adjustments** : Make adjustments due to the changes in the hardware



CD&A User Interface

ABi v810 Service - Confirmation, Diagnostics and Adjustments

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Main Recipes Production Options Services Virtual Live

Test List:

- Service Folder
 - Scheduled Tasks
 - System Communication Network Confirmation
 - Area Mode Image Confirmation
 - Camera Grayscale Adjustment
 - Camera Adjustment Confirmation
 - Camera Dark Confirmation
 - Camera Light Confirmation
 - X-ray Spot Adjustment
 - Stage Hysteresis Adjustment
 - System Offsets Adjustment
 - Magnification Adjustment
 - Base Image Quality Confirmation
 - System Image Quality Confirmation
 - High Mag Camera Grayscale Adjustment
 - High Mag Camera Adjustment Confirmation
 - High Mag Camera Dark Confirmation
 - High Mag Camera Light Confirmation
 - High Mag X-ray Spot Adjustment
 - High Mag Stage Hysteresis Adjustment
 - High Mag System Offsets Adjustment
 - High Magnification Adjustment
 - High Mag Base Image Quality Confirmation
 - High Mag System Image Quality Confirmation
 - X-ray Source Confirmation
 - Manual Tasks
 - Manual Confirmations
 - Fixed Rail Image Quality Confirmation
 - Adjustable Rail Image Quality Confirmation
 - High Mag Fixed Rail Image Quality Confirmation
 - High Mag Adjustable Rail Image Quality Confirmation
 - X-ray Cylinder Up/Down Confirmation
 - Panel Handling Confirmation
 - Motion Repeatability Confirmation
 - Manual Diagnostics
 - System Diagnostic
 - Manual Adjustments
 - X-ray Spot Deflection Adjustment
 - Manual Optical Confirmations
 - Optical Camera Light Confirmation
 - Manual Optical Adjustments
 - Optical System Fiducial Adjustment
 - Optical Calibration Adjustment
 - Optical Magnification
 - Optical Fringe Pattern Uniformity

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 - Optical Magnification
 - Optical Fringe Pattern Uniformity

Selected Tests

Run Continuously ☒ Run to First Failure

Message	Actions
---------	---------

v810 System XXL (PSP+VM) 801452

Run Selected Task with Independencies

The screenshot shows the 'v810 Service - Confirmation, Diagnostics and Adjustments' window. The interface includes a menu bar (File, Edit, Service, Tools, Actions, Window, Help), a toolbar, and a sidebar with icons for Main, Recipes, Production, Options, Services, and Virtual Live. The main area is divided into sections: 'Confirmation, Diagnostics, & Adjustments', 'Subsystem Status & Control', and 'X-ray Safety Test'. The 'Test List' section on the left is expanded, showing a hierarchy of tasks: 'Service Folder' (Scheduled Tasks, Manual Tasks), 'Manual Tasks' (Manual Confirmations, Manual Diagnostics, Manual Adjustments, Manual Optical Confirmations). A red box highlights the 'Testing task' section, which includes a list of tasks such as 'System Communication Network Conf', 'Area Mode Image Confirmation', 'Camera Grayscale Adjustment', 'Camera Adjustment Confirmation', 'Camera Dark Confirmation', 'Camera Light Confirmation', 'X-ray Spot Adjustment', 'Stage Hysteresis Adjustment', 'System Offsets Adjustment', 'Magnification Adjustment', 'Base Image Quality Confirmation', 'System Image Quality Confirmation', 'High Mag Camera Grayscale Adjustr', 'High Mag Camera Adjustment Confir', 'High Mag Camera Dark Confirmation', 'High Mag Camera Light Confirmation', 'High Mag X-ray Spot Adjustment', 'High Mag Stage Hysteresis Adjustme', 'High Mag System Offsets Adjustment', 'High Magnification Adjustment', 'High Mag Base Image Quality Confir', 'High Mag System Image Quality Conf', 'X-ray Source Confirmation', 'Fixed Rail Image Quality Confir', 'Adjustable Rail Image Quality Cor', 'High Mag Fixed Rail Image Quality', 'High Mag Adjustable Rail Image C', 'X-ray Cylinder Up/Down Confir', 'Panel Handling Confirmation', 'Motion Repeatability Confirmation', 'System Diagnostic', 'X-ray Spot Deflection Adjustment', and 'Manual Optical Confirmations'. The 'Run Selected Tests' button is highlighted with a red box and the text 'Click to Run Selected Tests'. The 'Loop Count' field is set to 1, and the 'Run to First Failure' checkbox is checked. The 'Loop Continuously' checkbox is unchecked. The 'Run the test until first error message encountered' text is also present. The 'No of loop for particular test' text is also present. The 'Looping the test continuously until error prompted out / manually abort the test' text is also present.

Testing task

Click to Run Selected Tests

Run the test until first error message encountered

Loop Count: 1

Loop Continuously

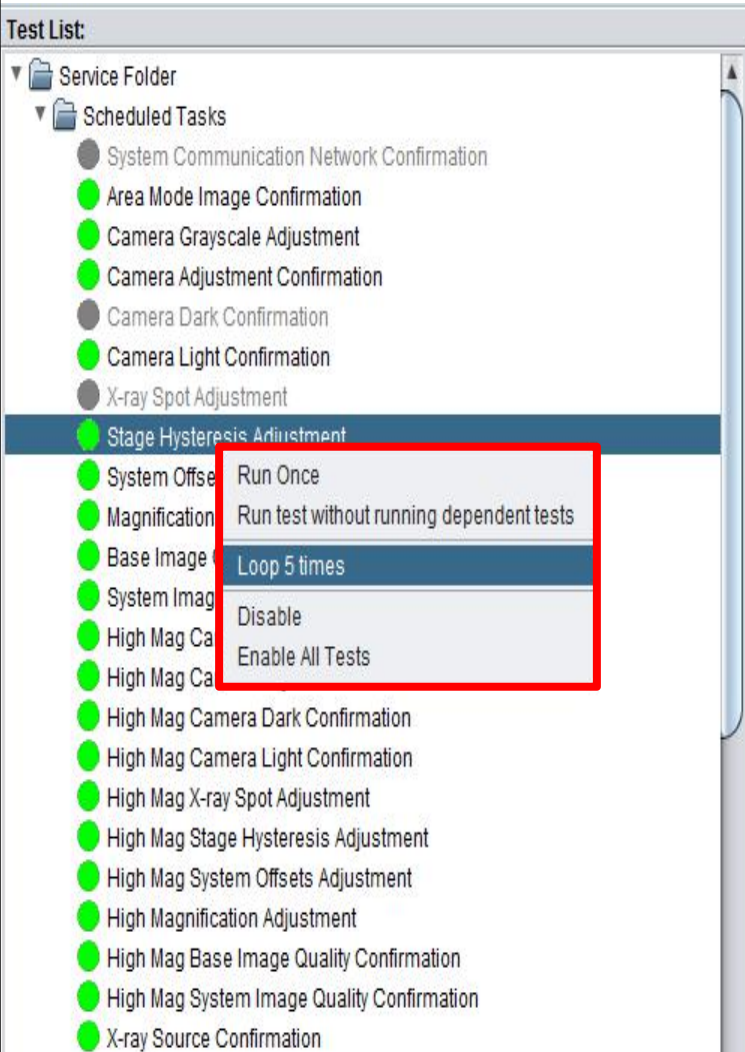
Run to First Failure

No of loop for particular test

Looping the test continuously until error prompted out / manually abort the test



CD&A Test Option



Run Once: Will run all the test before the selected task.

Run test without dependent tests : Will only run the task selected by user without running the prior test.

Loop 5 times : Will run the test according to the looping number provided inside the **Loop Count**

Loop Count:

Disable: Skip / Ignore the test and will run other test.

: The icon will turn grey

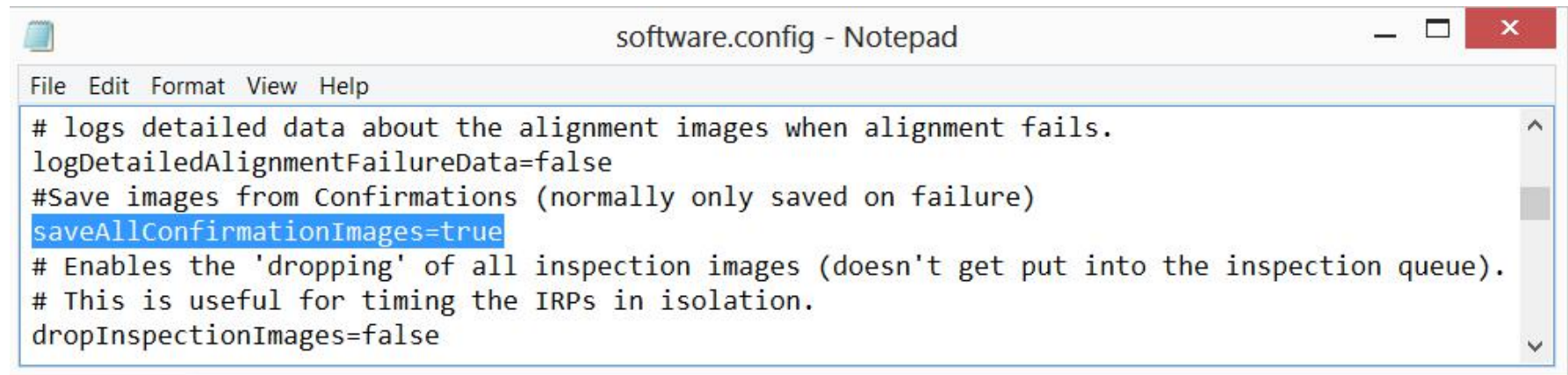
Enable All Tests: Will enable all the test to be tested

- **Hardware files**

- C:\Program Files\AXI System\V810\<Release#>\calib\hardware.calib
- C:\Program Files\AXI System\V810\<Release#>\config\hardware.config

- **Software files**

- C:\Program Files\AXI System\V810\<Release#>\config\software.config
- To view log images/ data must turn on the logging in the software.config file search for “saveAllConfirmationImages=false” to “saveAllConfirmationImages=true”

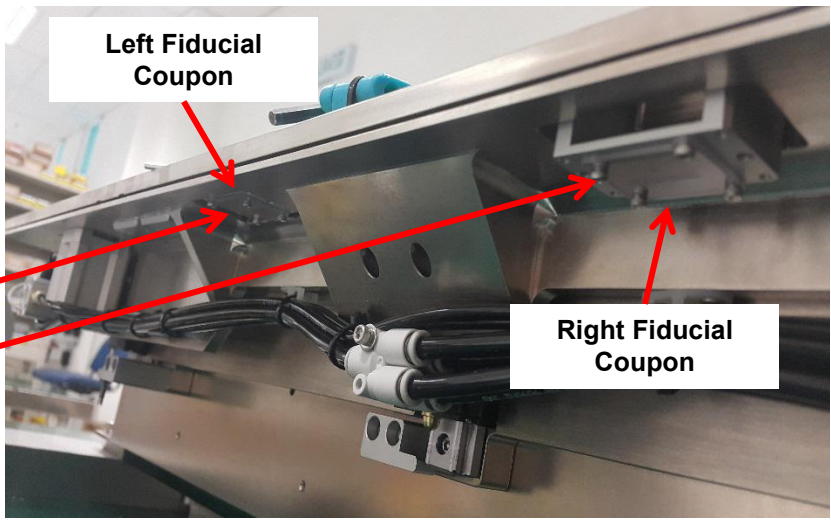
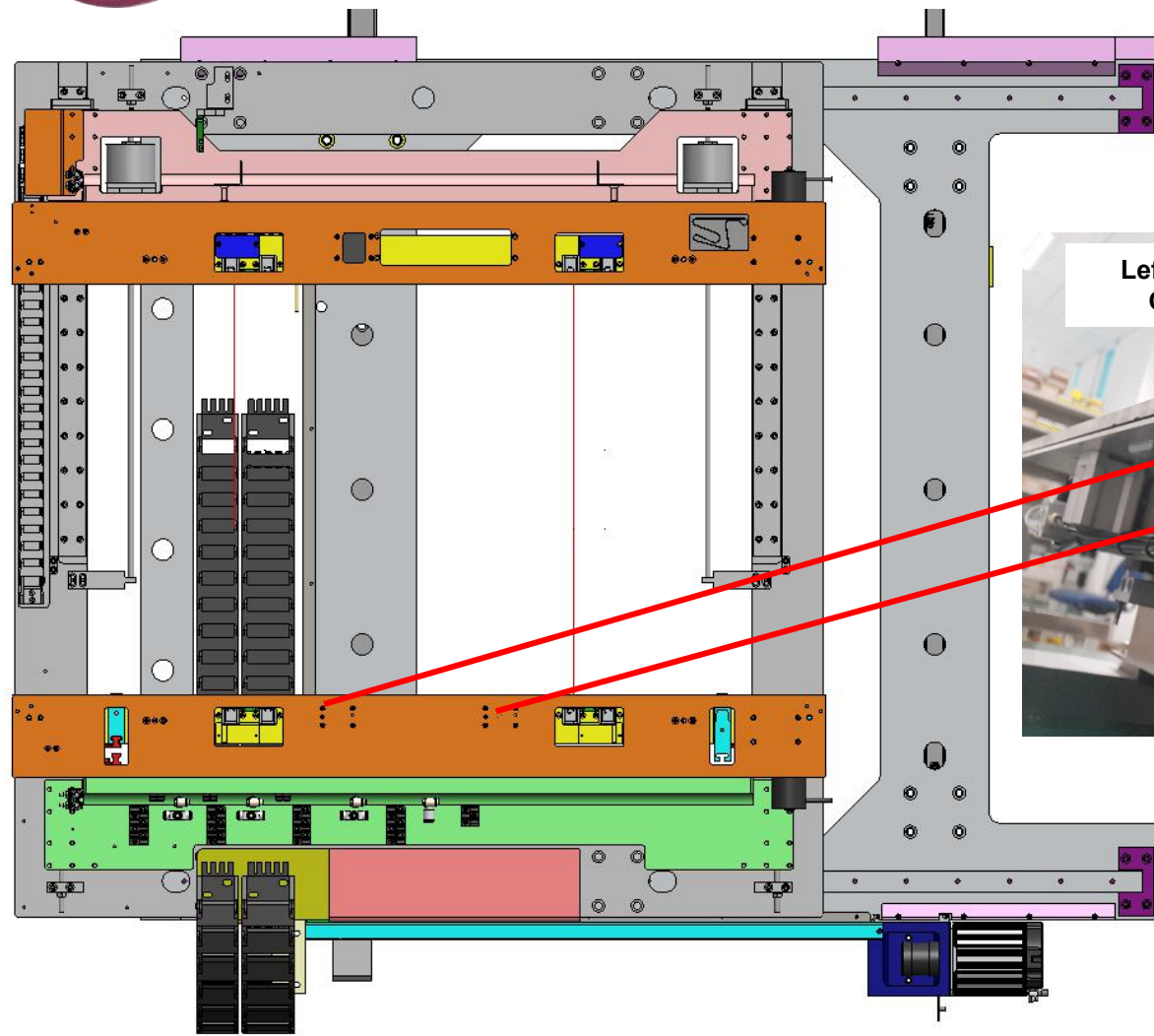


```
File Edit Format View Help
# logs detailed data about the alignment images when alignment fails.
logDetailedAlignmentFailureData=false
#Save images from Confirmations (normally only saved on failure)
saveAllConfirmationImages=true
# Enables the 'dropping' of all inspection images (doesn't get put into the inspection queue).
# This is useful for timing the IRPs in isolation.
dropInspectionImages=false
```

- **Save the file, restart the V810 software to enable the logging of the log images**
- **Verify all the latest log files with latest date modify**



Fiducial Coupon Location





Fiducial Coupon

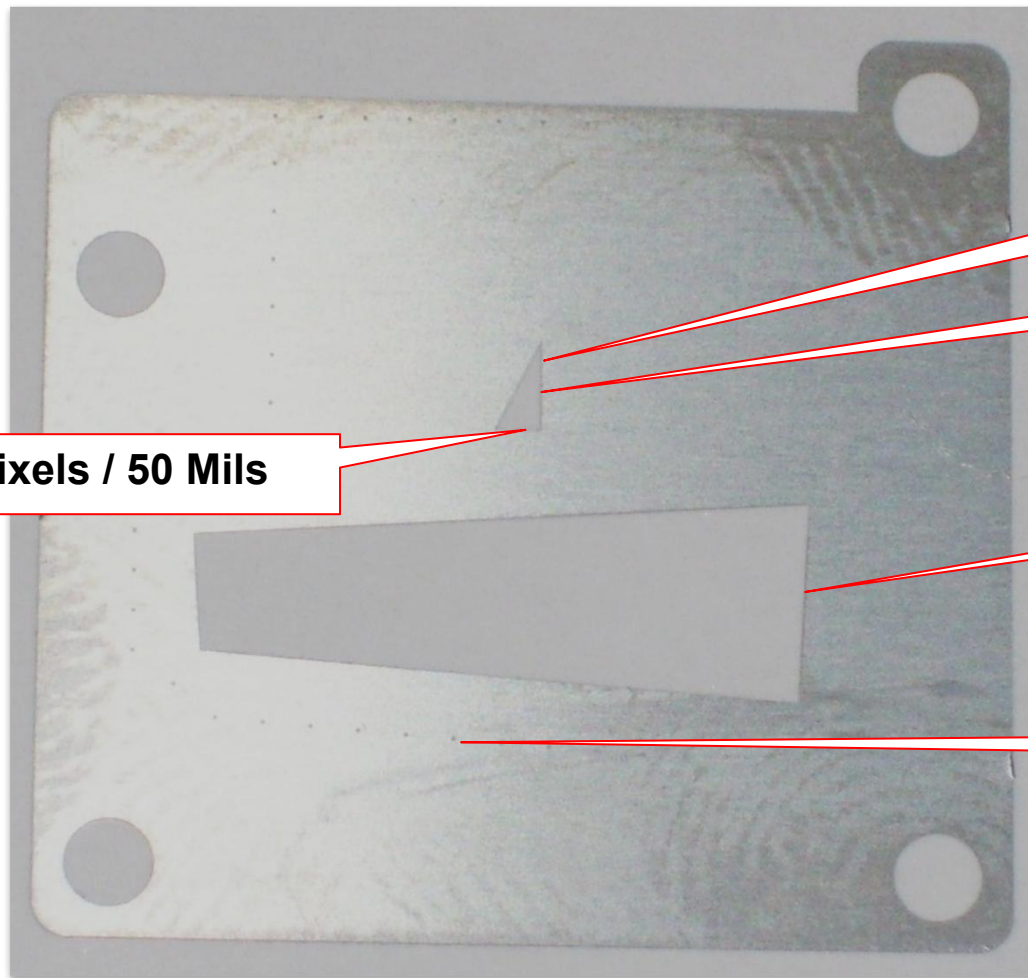
160 Pixels / 100 Mils

System Fiducial

80 Pixels / 50 Mils

Quadrilateral

Small holes



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 - High Mag Camera Dark Confirmation
 - High Mag Camera Light Confirmation
 - High Mag X-ray Spot Adjustment
 - High Mag Stage Hysteresis Adjustment
 - High Mag System Offsets Adjustment
 - High Magnification Adjustment
 - High Mag Base Image Quality Confirmation
 - High Mag System Image Quality Confirmation
 - X-ray Source Confirmation

1. Checks for network communication from the system controller to the IRPs.

2. To check for the network connectivity between devices below with the main controller:

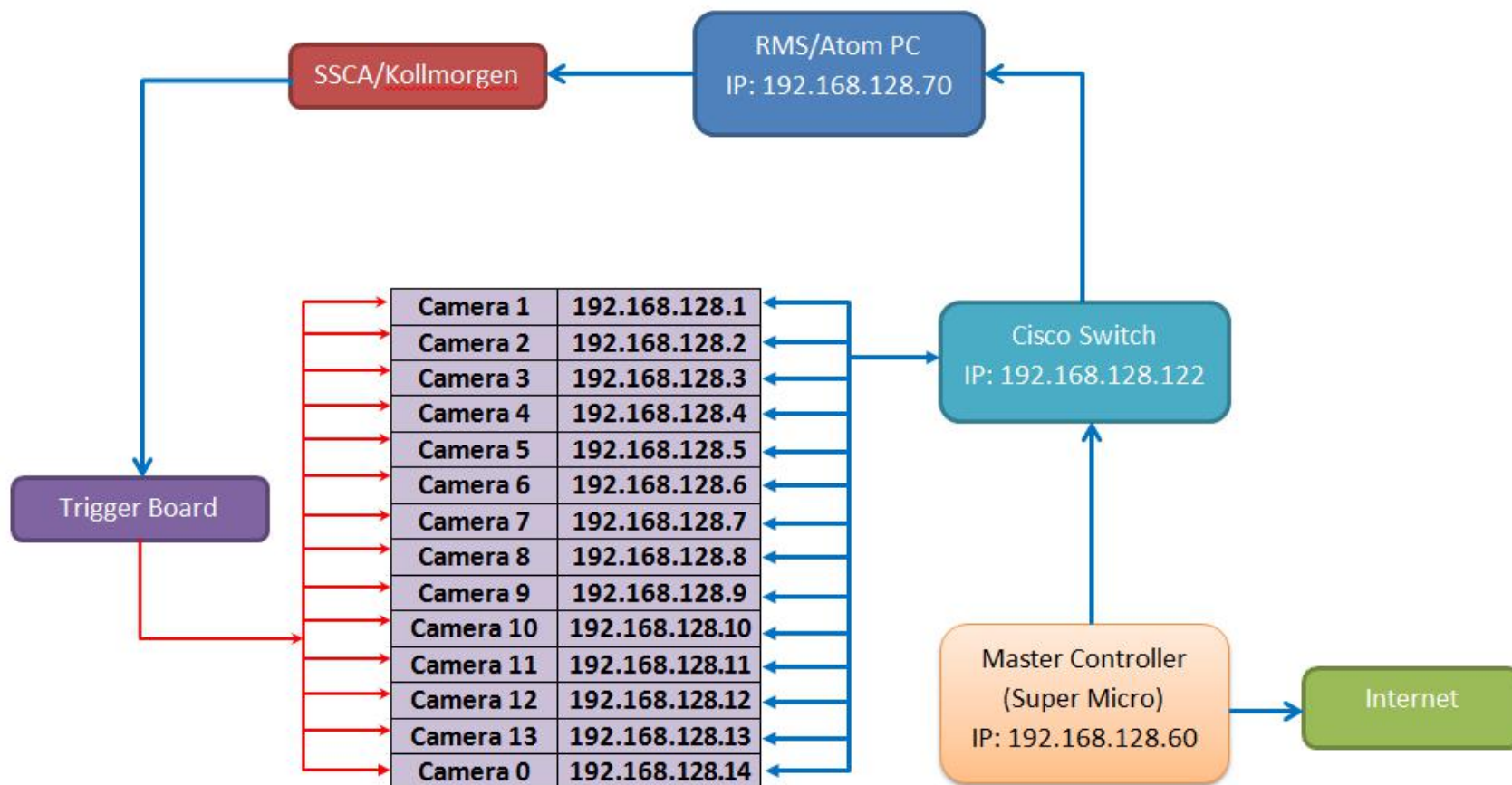
- 4 IRPs,
- 14 X-Ray cameras
- Gigabit Switch
- Atom PC [RMS]
- SSCA Controller

This is a very simple “ping” test that assures network connectivity exists between all the components in the V810 system. Each specific subsystem also has connection tests that occur when the specific subsystems are initialized (ie. When the first connection to the cameras is initiated).

3. Runs at every time machine “Start-up”.



System Communication Network Confirmation





System Communication Network Confirmation

If communication to one or more devices fails:

1. Check Power : Look for LED light on the failing devices
2. Check Cabling: Reseat the cables or check for any damaged cables
3. Power cycle the device
4. Check network configuration inside the master controller and also in the hardware.config
5. Using command prompt, try to ping each devices and check whether there is a feedback from each device.

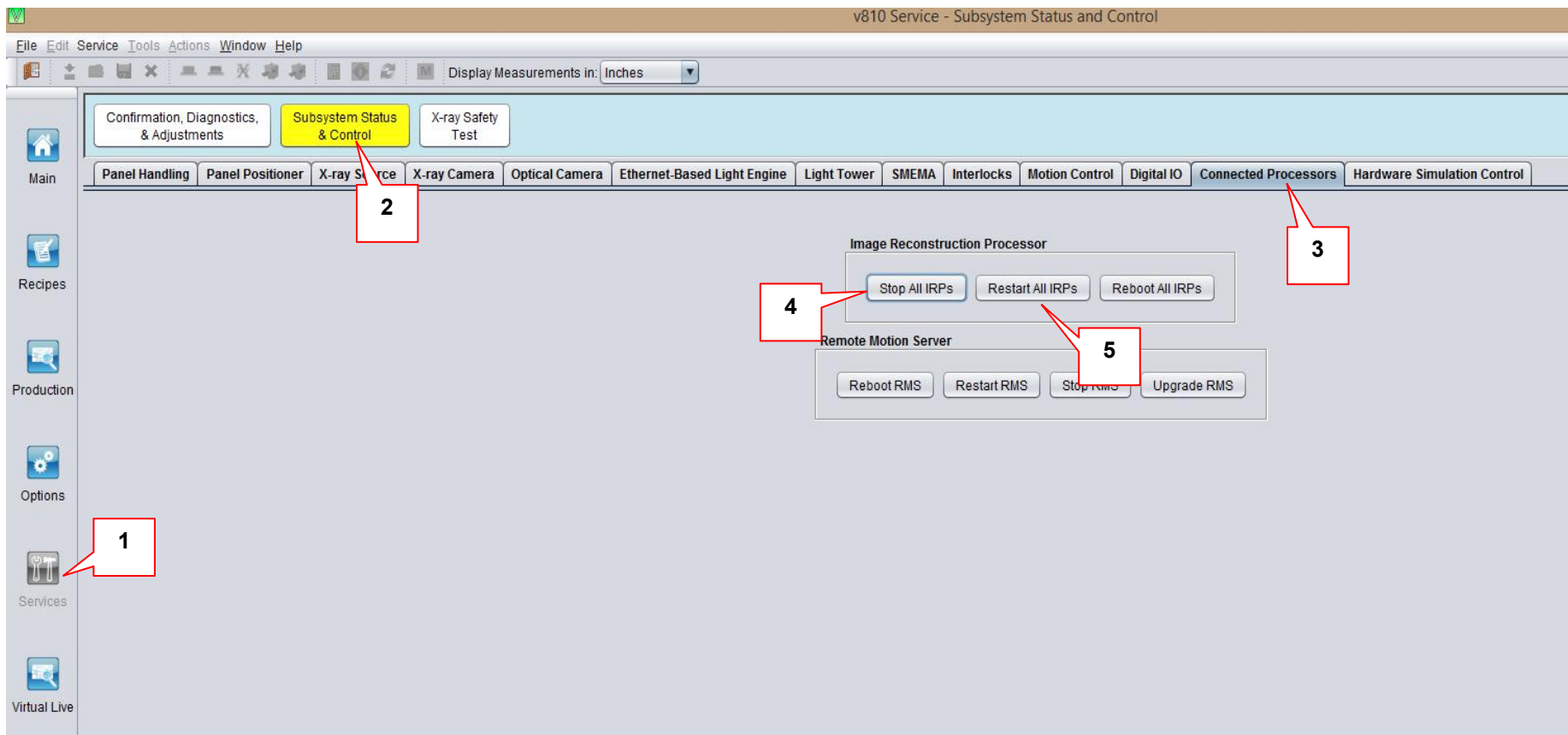


IRP Communication Error

During start-up: Error prompt out regarding communication error with IRP

Possible root cause: IRP Software not enable to start up

Solution: Start-up IRP SW with V810 GUI





IRP Communication Error

Check on the task manager, ensure at Details tab, the IRP.exe is running (8 IRP's)

Name	PID	Status	User name	CPU	Memory (p...	Description
3CTftpSvc.exe	1616	Running	SYSTEM	00	516 K	3CTftpSvc
armsvc.exe	1636	Running	SYSTEM	00	680 K	Adobe Acrobat Update Service
audiodg.exe	20336	Running	LOCAL SE...	00	3,312 K	Windows Audio Device Graph Isolation
cmd.exe	48240	Running	Administr...	00	424 K	Windows Command Processor
conhost.exe	8532	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	740	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	59120	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	59728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	28372	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	50664	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	25744	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	5728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	44272	Running	Administr...	00	532 K	Console Window Host
csrss.exe	576	Running	SYSTEM	00	1,468 K	Client Server Runtime Process
csrss.exe	71412	Running	SYSTEM	00	1,136 K	Client Server Runtime Process
dwm.exe	26620	Running	DWM-1	00	9,352 K	Desktop Window Manager
explorer.exe	29504	Running	Administr...	00	37,328 K	Windows Explorer
IPROSetMonitor.exe	1680	Running	SYSTEM	00	608 K	Intel® PROSet Monitoring Service
IRP.exe	5408	Running	SYSTEM	00	624 K	IRP
IRP.exe	38272	Running	SYSTEM	00	620 K	IRP
IRP.exe	65424	Running	SYSTEM	00	628 K	IRP
IRP.exe	50904	Running	SYSTEM	00	620 K	IRP
IRP.exe	4948	Running	SYSTEM	00	632 K	IRP
IRP.exe	24496	Running	SYSTEM	00	628 K	IRP
IRP.exe	25536	Running	SYSTEM	00	620 K	IRP
IRP.exe	10964	Running	SYSTEM	00	632 K	IRP
javaw.exe	10440	Running	Administr...	00	260,716 K	Java(TM) Platform SE Binary

8 IRP.exe should
run simultaneously



IRP Communication Error

Check for service task manager, 10 tasks below must run automatically

Name	PID	Status	User name	CPU	Memory (p...	Description
3CTftpSvc.exe	1616	Running	SYSTEM	00	516 K	3CTftpSvc
armsvc.exe	1636	Running	SYSTEM	00	680 K	Adobe Acrobat Update Service
audiodg.exe	20336	Running	LOCAL SE...	00	3,312 K	Windows Audio Device Graph Isolation
cmd.exe	48240	Running	Administr...	00	424 K	Windows Command Processor
conhost.exe	8532	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	740	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	59120	Running	SYSTEM	00	380 K	Console Window Host
conhost.exe	59728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	28372	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	50664	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	25744	Running	SYSTEM	00	384 K	Console Window Host
conhost.exe	5728	Running	SYSTEM	00	388 K	Console Window Host
conhost.exe	44272	Running	Administr...	00	532 K	Console Window Host
csrss.exe	576	Running	SYSTEM	00	1,468 K	Client Server Runtime Process
csrss.exe	71412	Running	SYSTEM	00	1,136 K	Client Server Runtime Process
dwm.exe	26620	Running	DWM-1	00	9,352 K	Desktop Window Manager
explorer.exe	29504	Running	Administr...	00	37,328 K	Windows Explorer
IPROSetMonitor.exe	1680	Running	SYSTEM	00	608 K	Intel® PROSet Monitoring Service
IRP.exe	5408	Running	SYSTEM	00	624 K	IRP
IRP.exe	38272	Running	SYSTEM	00	620 K	IRP
IRP.exe	65424	Running	SYSTEM	00	628 K	IRP
IRP.exe	50904	Running	SYSTEM	00	620 K	IRP
IRP.exe	4948	Running	SYSTEM	00	632 K	IRP
IRP.exe	24496	Running	SYSTEM	00	628 K	IRP
IRP.exe	25536	Running	SYSTEM	00	620 K	IRP
IRP.exe	10964	Running	SYSTEM	00	632 K	IRP
java.exe	10440	Running	Administr...	00	200,716 K	Java(TM) Platform SE binary
lsass.exe	764	Running	SYSTEM	00	7,084 K	Local Security Authority Process
nvsvsc.exe	988	Running	SYSTEM	00	1,716 K	NVIDIA Driver Helper Service, Version 347.25
nvsvsc.exe	21568	Running	SYSTEM	00	3,028 K	NVIDIA Driver Helper Service, Version 347.25
nvwmif4.exe	148	Running	SYSTEM	00	680 K	NVIDIA WMI Provider
nvwmif4.exe	39840	Running	SYSTEM	00	3,204 K	NVIDIA WMI Provider
nvxdsync.exe	29620	Running	SYSTEM	00	4,700 K	NVIDIA User Experience Driver Component
perl.exe	4516	Running	Administr...	00	692 K	perl
ReconstructionServi...	1784	Running	SYSTEM	00	3,584 K	ReconstructionService
SearchIndexer.exe	1820	Running	SYSTEM	00	30,804 K	Microsoft Windows Search Indexer
services.exe	752	Running	SYSTEM	00	4,304 K	Services and Controller app
smss.exe	436	Running	SYSTEM	00	216 K	Windows Session Manager
spoolsv.exe	1468	Running	SYSTEM	00	2,240 K	Spooler SubSystem App
svchost.exe	832	Running	SYSTEM	00	6,184 K	Host Process for Windows Services
svchost.exe	880	Running	NETWORK...	00	6,964 K	Host Process for Windows Services
svchost.exe	468	Running	LOCAL SE...	00	13,084 K	Host Process for Windows Services
svchost.exe	804	Running	SYSTEM	00	26,924 K	Host Process for Windows Services

10 task that should be run automatically:

1. 3CTftpSvc.exe

2. IRP.exe (8 task)

3. ReconstructionService.exe



System Communication Network Confirmation

If found any of the task is not running, please manually run it from the directory below:

- 3CTftpSvc.exe : C:\Program Files\AXI System\v810\<REV>\service
- ReconstructionService.exe : C:\Program Files\AXI System\v810\<REV>\service
- Irp.exe : Restart all IRP's at V810 GUI

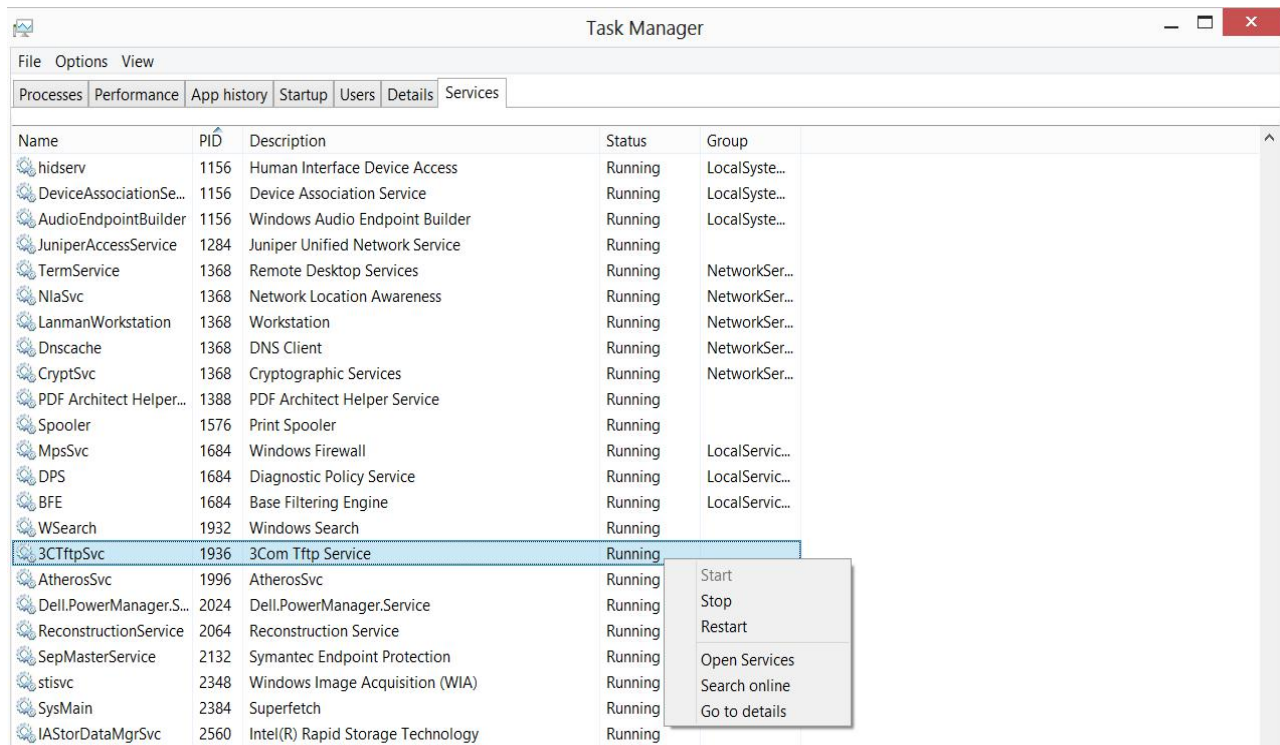


IRP Communication Error

At task manager, go to the Services tab. Ensure the following task is running:

- 1. 3CTfrpSvc.exe
- 2. ReconstructionService.exe
- 3. If the task not running, manually turn it on by :

Select the task-> Right Click -> Start

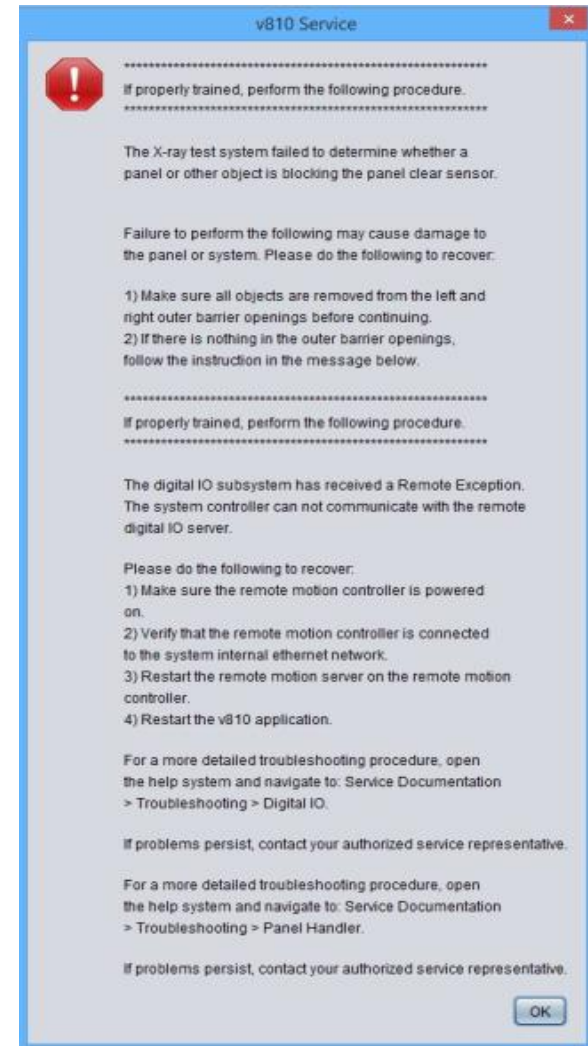
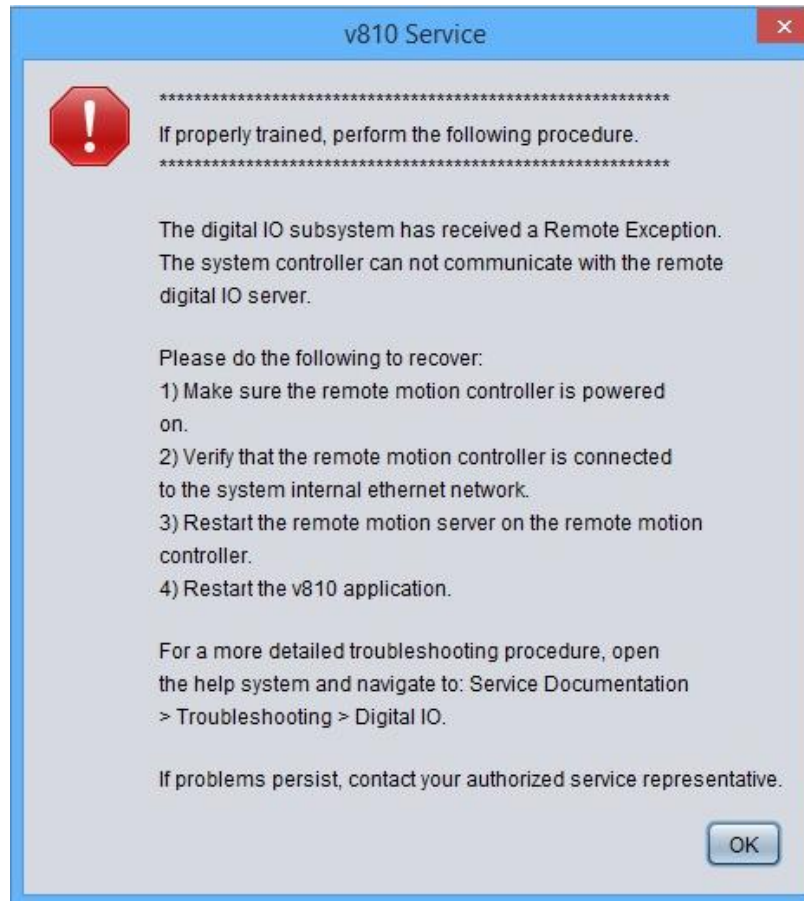




RMS Communication Error

Remote Motion Server (RMS) failed at Start-Up

Sample error message





RMS Communication Error

1. Using CMD, insert “ping 192.168.128.70”

Expected to see return from the Atom PC as shown with the TTL=128ms, else the connection between Master Controller and Atom PC might be lost.

```
Administrator: Command Prompt

Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

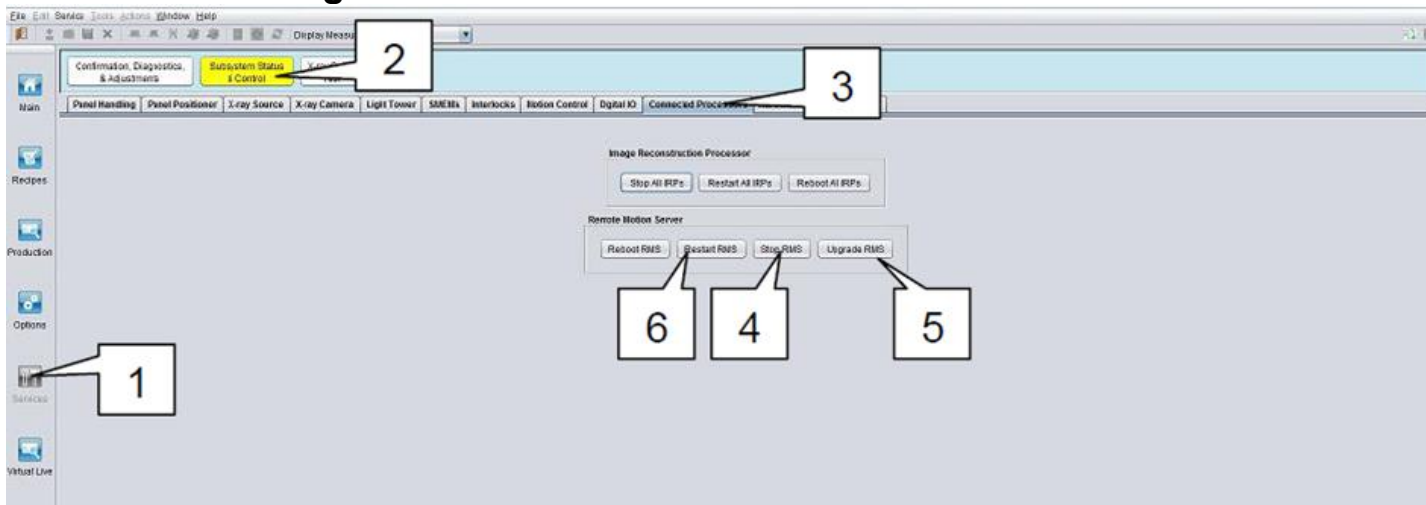
C:\Users\Administrator>ping 192.168.128.70

Pinging 192.168.128.70 with 32 bytes of data:
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128
Reply from 192.168.128.70: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.128.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>
```

2. Restart RMS using V810





RMS Communication Error

3. Remote Desktop into Atom PC using the following credentials:

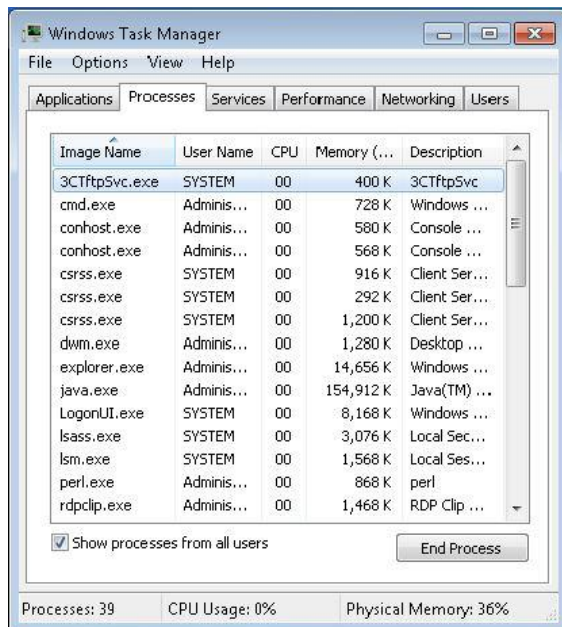
Computer: 192.168.128.70

Username: Administrator

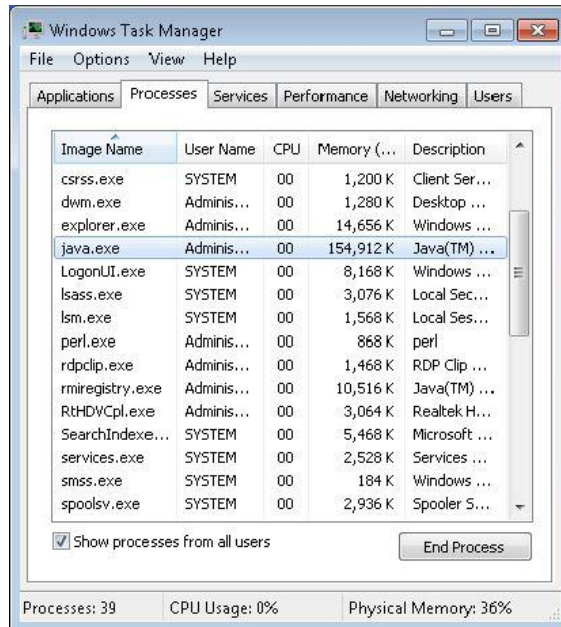
Password: Please!



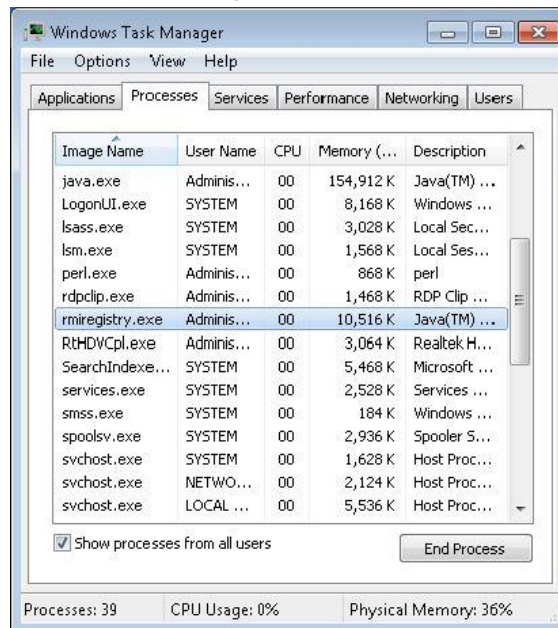
4. Go to the Task Manager and Processes tab, 3 task should be run automatically as shown below



3CTftpSvc.exe



java.exe



rmiregistry.exe



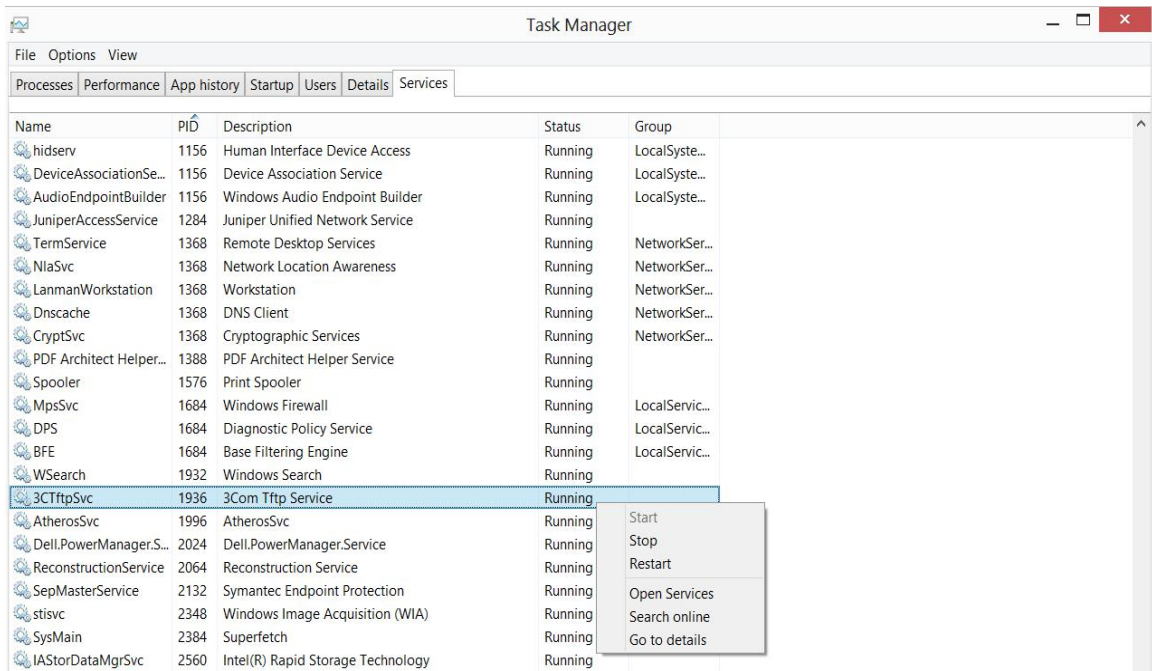
RMS Communication Error

If found any of it is not running, manually run it from directory below (Atom PC)

- 3CTftpSvc.exe : C:\Program Files\AXI System\v810\<Rev>\bin
- rmiregistry.exe : C:\Program Files\AXI System\v810\<Rev>\jre\1.7.0_x86\bin
- Java.exe : will auto run every time restart RMS at GUI

At task manager, go to the Services tab. Ensure the following task is running: 3CTftpSvc.exe

If the task not running, manually turn it on by : Select the task-> Right Click -> Start



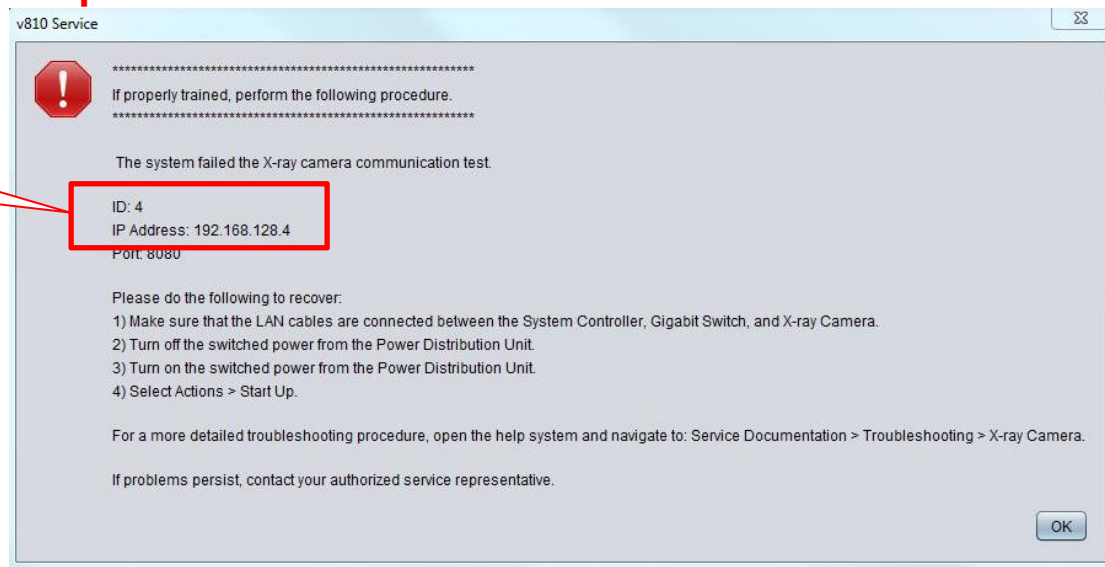


X-Ray Camera Communication Error

Camera fail at Start-up:

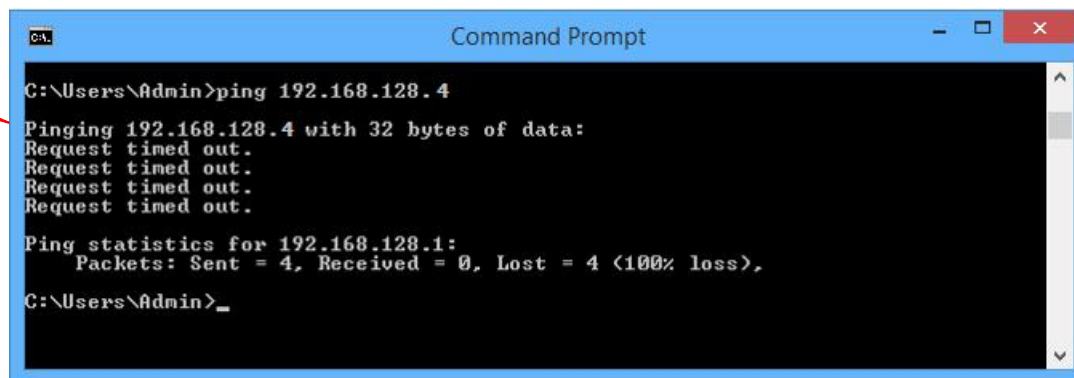
Error Message:

Camera ID and
IP Address



Action: Manually ping the IP Address of the camera for communication test

Manually ping the respective IP
Address

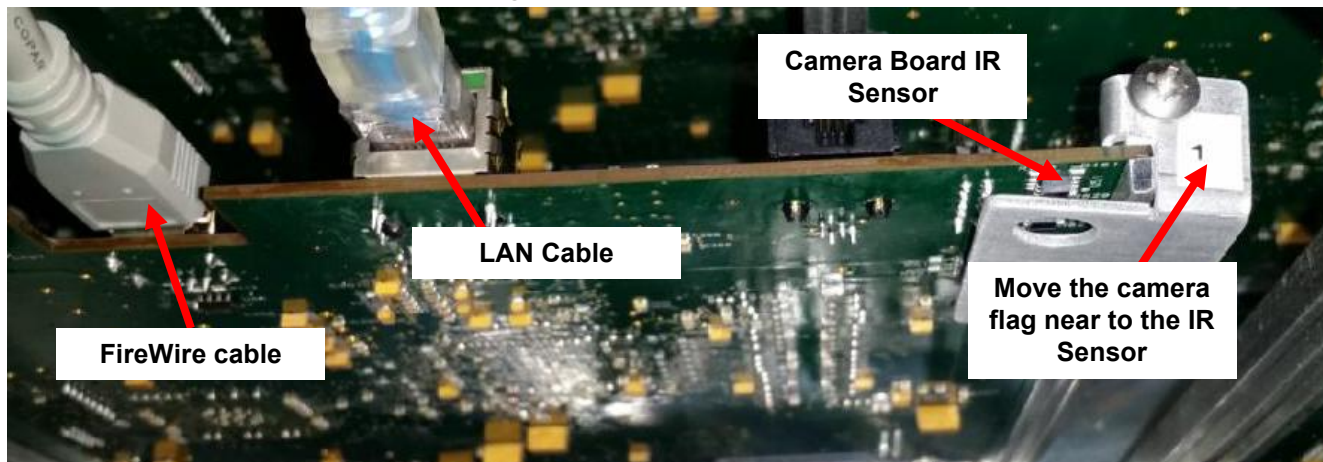




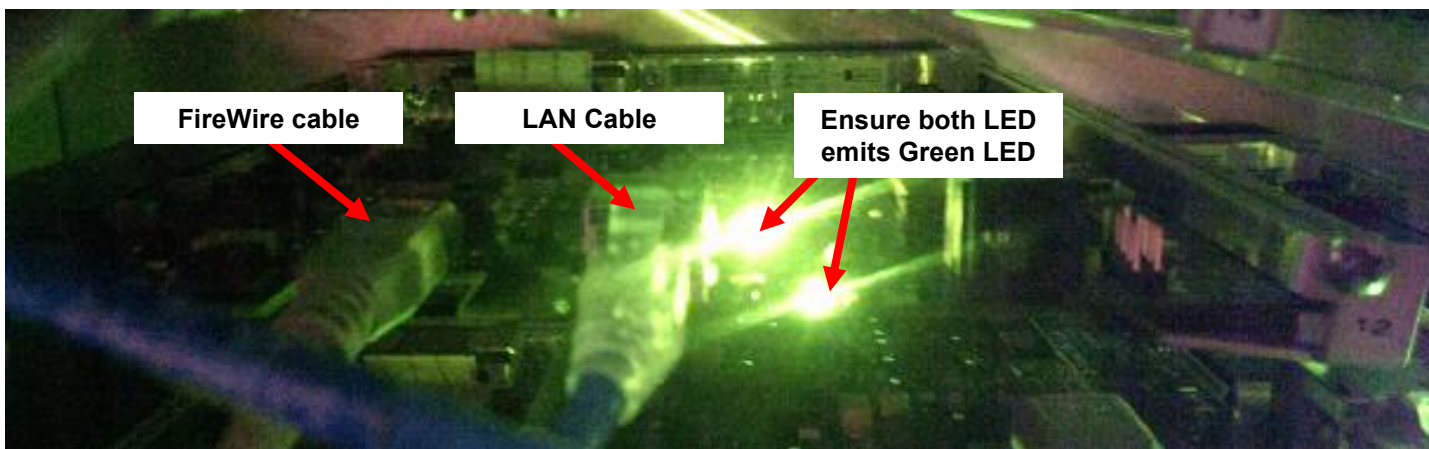
System Communication Network Confirmation

If manually ping the camera fail, perform the following steps to recover the camera communication:

- 1) Ensure the respective Camera Flag ID is near to the sensor of the camera board



- 2) Re-connect LAN cable and the FireWire cable at the camera. Ensure both LED emits Green.

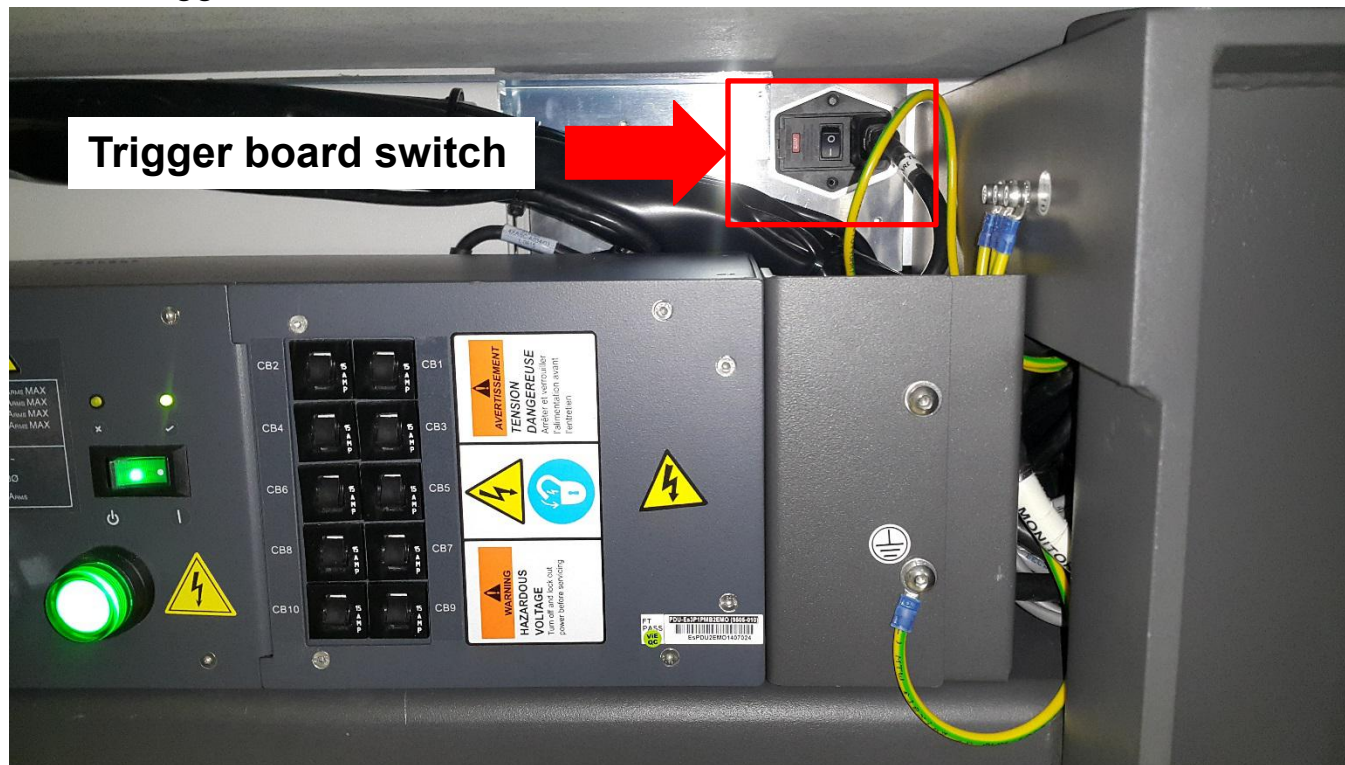




System Communication Network Confirmation

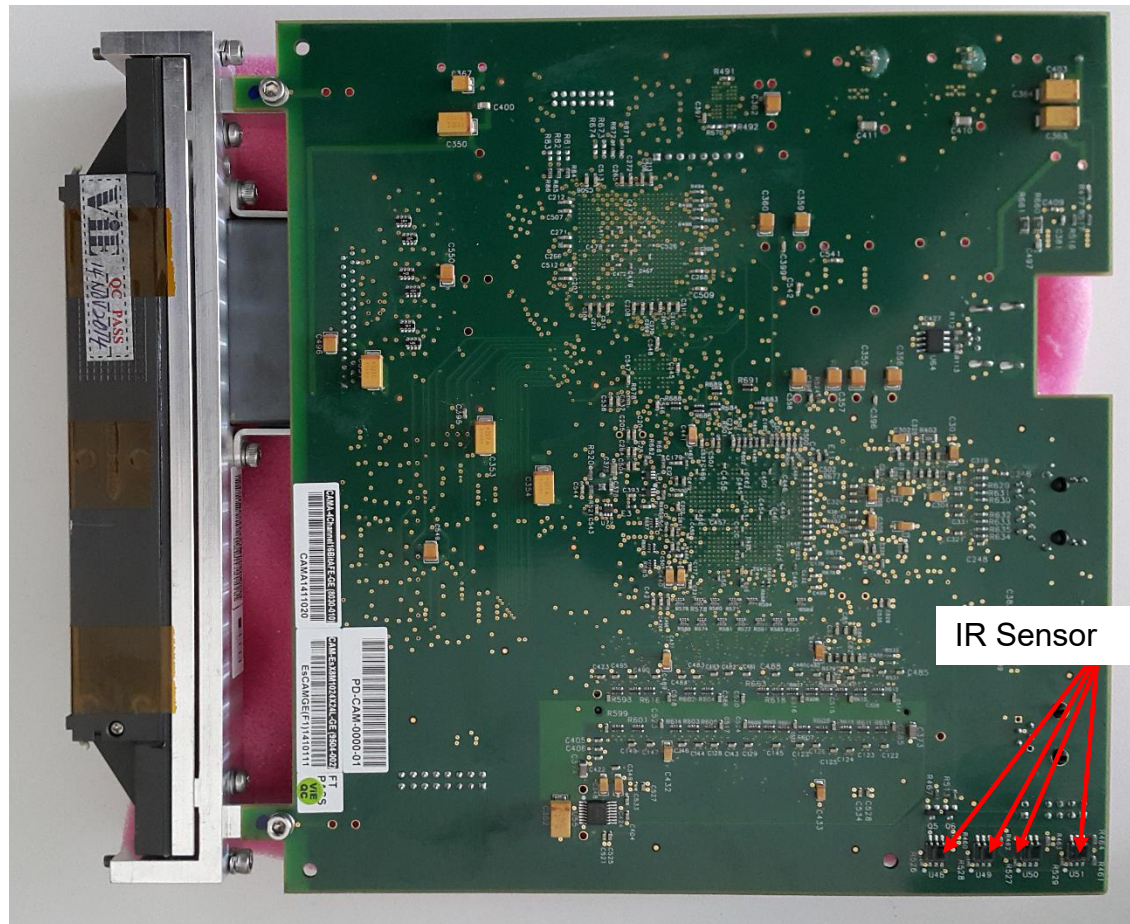
3) Power cycle the trigger board (Off for 30 sec and on back)

Remarks: The trigger board switch is located behind the PDU



4) Open CMD and ping the respective camera again. If fail, try to clean the IR Sensor

- 5) Clean the IR Sensor with IPA for the faulty camera. Reposition back the camera flag near the IR Sensor and try to manually ping back the camera using the CMD



Area Mode Image Confirmation

Test List:

- ▼ Service Folder
 - ▼ Scheduled Tasks
 - System Communication Network Confirmation
 - Area Mode Image Confirmation
 - Camera Grayscale Adjustment
 - Camera Adjustment Confirmation
 - Camera Dark Confirmation
 - Camera Light Confirmation
 - X-ray Spot Adjustment
 - Stage Hysteresis Adjustment
 - System Offsets Adjustment
 - Magnification Adjustment
 - Base Image Quality Confirmation
 - System Image Quality Confirmation
 - High Mag Camera Grayscale Adjustment
 - High Mag Camera Adjustment Confirmation
 - High Mag Camera Dark Confirmation
 - High Mag Camera Light Confirmation
 - High Mag X-ray Spot Adjustment
 - High Mag Stage Hysteresis Adjustment
 - High Mag System Offsets Adjustment
 - High Magnification Adjustment
 - High Mag Base Image Quality Confirmation
 - High Mag System Image Quality Confirmation
 - X-ray Source Confirmation

Purpose : To verify all 14 cameras quality by capturing snapshot picture

: Make sure there is no debris on the camera

: Make sure no bad segment on the camera's scintillator

Frequency : Runs at system Start-up if the machine idle more than 24 hours

Method : Stage move to the center of the machine

: AWA fully open (19 inch)

: Inner barrier open

: X-Ray source turn on

: Image captured (Uses synthetic trigger to capture image)

Results : Log result to the

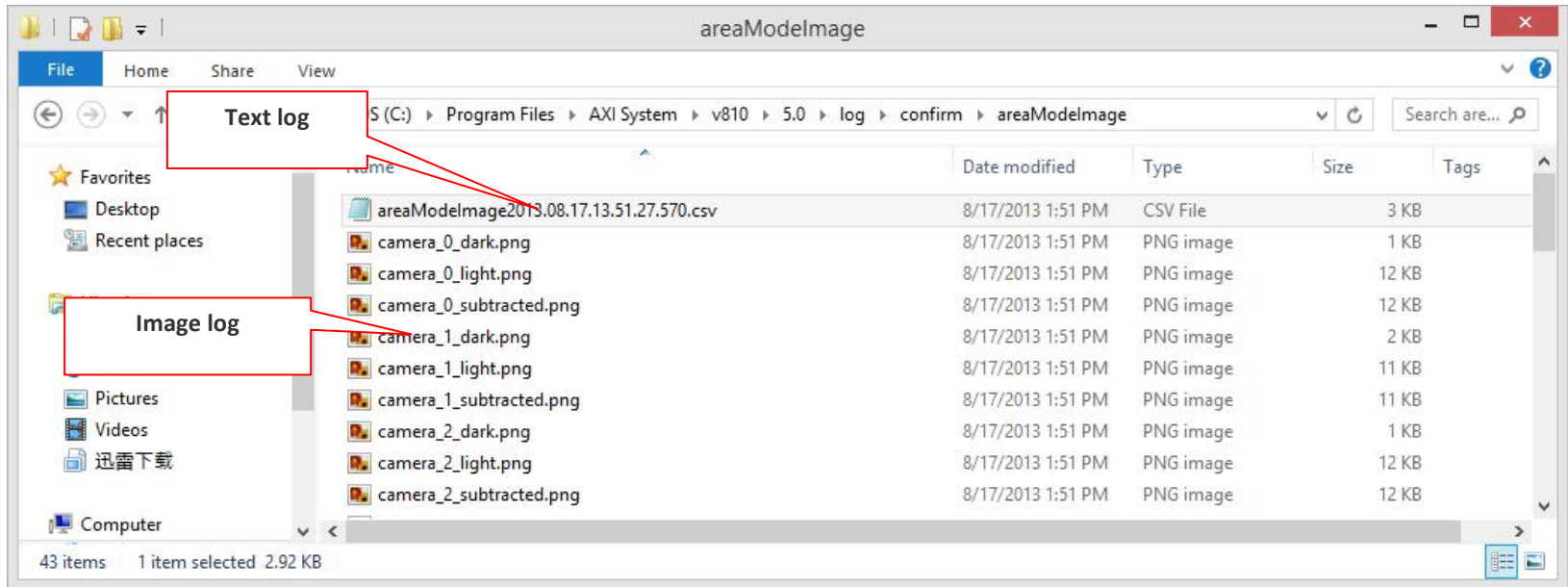
C:\Program Files\AXI System\v810\<REV>\log\confirm\areamodelImage

Remarks : No board inside the machine during the area mode image test else will drop inside the machine during system running CD&A task



Area Mode Image Confirmation

Logs result to C:\Program Files\AXI System\v810\<REV>\log\confirm\areamodelImage



Each time calibration will generate 42 image log files and 1 text log file.

3 image logs for each camera with 1 text log (14 camera's)

- camera_xx_dark.png (must all dark image)
- camera_xx_subtracted.png (Grey but darker than camera light image)
- camera_xx_light.png (Grey but light than camera subtracted image)
- areaModelImageYYYY.MM.DD.HH.MM.SS.XXX

Sort log file with date modified (descending) to review the latest result.



Area Mode Image Troubleshooting



Sample Bad Segment

Error 1: Bad Segment

Troubleshooting:

1. Check for any debris on any foreign material on the top of Field Stop Assembly
2. Verify the inner can open and close manually using Digital IO

Error 2: Fail Communication with insufficient Jumbo Frame Size

Troubleshooting:

1. Check the flag for all the camera's
2. Reseat / check the Cat5e cable
3. Ping all the camera and expect to see the return of the TTL=64 for each camera
4. If multiple camera failed, replace the trigger board



Area Mode Image Troubleshooting

Error 3: Camera 0,1,...13 fail at Area mode image confirmation / Camera greyscale adjustment

Troubleshooting:

1. In software.config change “saveAllConfirmationImages=false” to “saveAllConfirmationImages=true”
2. Close and re-open the V810 software
3. Delete all images at area mode image confirmation folder
4. Run CD&A
5. Check each image for the failure camera segment
6. Replace new camera
7. Upgrade camera firmware
8. Re-run CD&A



Area Mode Image Troubleshooting

Error 4: Area mode image confirmation test stuck at 0 %

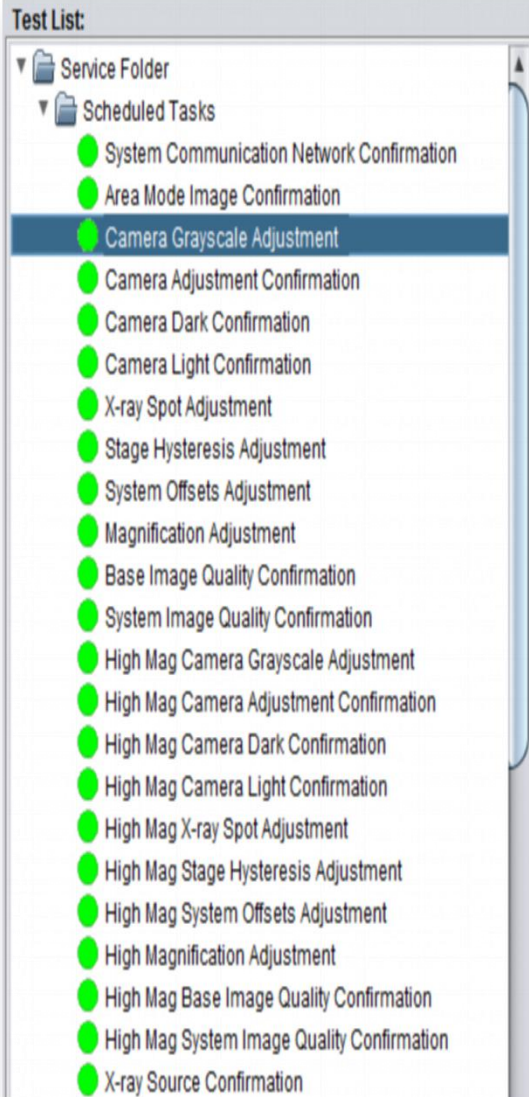
Root Cause: X-ray source not warm up completely or overload

Troubleshooting:

1. Go to X-Ray source tab at V810 software and initialize
2. Try to warm up the tube using the V810 software. If the software unable to warm up the tube, then need to manually warm up the tube using the MFX software.
3. Close the V810 software, open the MFX Software.
4. Try to manually warm up the tube using under this 3 conditions:
 - a. Tube Voltage: 80kV ; Tube current : 100 μ A ; Focus Mode: Small (After 1 hour turn off)
 - b. Tube Voltage: 110kV ; Tube current : 146 μ A ; Focus Mode : Middle off) (After 1 hour turn off)
 - c. Tube Voltage: 130kV ; Tube current : 200 μ A ; Focus Mode : Large off) (After 1 hour turn off)
5. Turn of the X-Ray, click “Tool” and press disconnect before exit the MFX software.
6. Open V810 software, and warm-up the tube
7. Run CD&A

If during ramping up, the overload icon is blinking, the tube is overload unstable and requires replacement

Camera Grayscale Adjustment



- Purpose** : It's a self-calibration on segment, pixel gains and offset for each camera
- Frequency** : Runs at system Start-up if the machine idle more than 2 hours
: During loading and unloading if more than 2 hours
- Results** : Log result to the
C:\Program Files\AXI System\v810\5.0\log\calib\CameraGrayscaleLowMag
C:\Program Files\AXI System\v810\5.0\log\calib\CameraGrayscaleHighMag
- Method** : Move the stage to the center (AWA width > 4 inch)
: Move the stage to the minimum Y-axis hard stop (AWA width < 4 inch)
: Self calibration on segment and pixel gain and offset for each camera
: Use synthetic triggers and area mode images
- Remarks** : Camera Grayscale Adjustment occurs in 4 stages currently
- Measure segmentOffset
 - SegmentGain
 - PixelOffset
 - MeasurePixelGain



Camera Grayscale Adjustment

The segment gain and offset calibrate the analog gains on the front end, these adjustments are done in the A/D converters themselves and each CCD has its own setting.

The Pixel gain and offset are all digital correction.

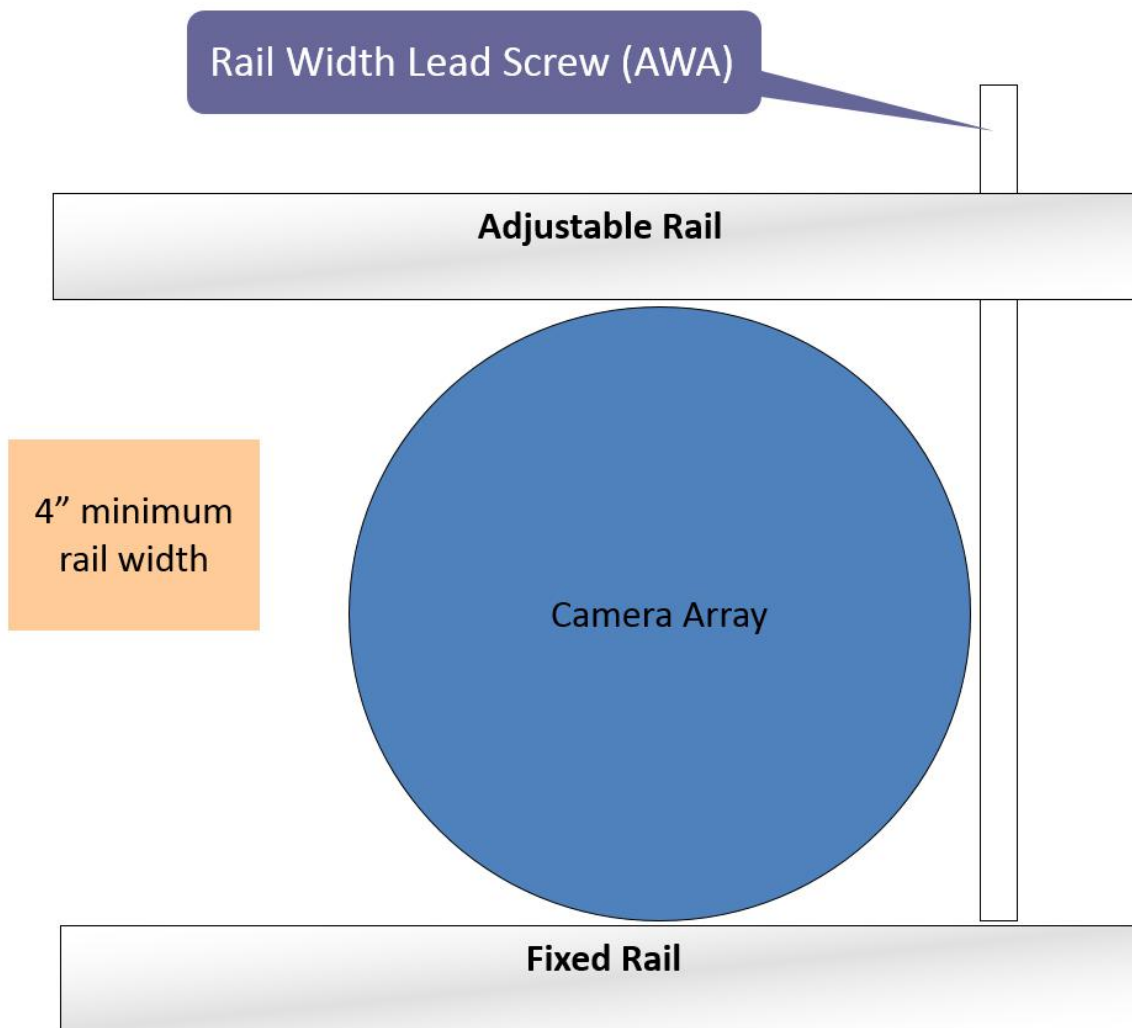
Black $\approx$ 6 and White $\approx$ 250

Adjustment fails only if the hardware is unable to adjust the contrast & brightness for proper pixel and segment gray level.

Logs results to:

- i. C:\Program Files\AXI System\v810\5.0\log\calib\CameraGrayscaleLowMag
- ii. C:\Program Files\AXI System\v810\5.0\log\calib\CameraGrayscaleHighMag

Camera Grayscale Adjustment





Camera Grayscale Adjustment Troubleshooting

Confirmation, Diagnostics, & Adjustments

Subsystem Status & Control

X-ray Safety Test

Test List:

- Service Folder
 - Scheduled Tasks
 - System Communication Network Confirmation
 - Area Mode Image Confirmation
 - Camera Grayscale Adjustment
 - Camera Adjustment Confirmation
 - Camera Dark Confirmation
 - Camera Light Confirmation
 - X-ray Spot Adjustment
 - Stage Hysteresis Adjustment
 - System Offsets Adjustment
 - Magnification Adjustment
 - Base Image Quality Confirmation
 - System Image Quality Confirmation
 - X-ray Source Confirmation

Run Selected Tests

Loop Count: 1

Loop Continuously

Run to First Failure

#	Name	Status	Message	Actions
1	Connection to IRPs Confirmation	Passed		
2	Connection to RMS Confirmation	Passed		
3	Area Mode Image Confirmation	Passed		
4	Camera Grayscale Adjustment	FAILED	Camera(s) 7 FAILED*****If proper... Check for debris on camera array.	

AXI System v810

!

.....

If properly trained, perform the following procedure.

.....

The Camera Grayscale Adjustment FAILED

Camera(s) 13,12,11,10,9,8,7,6,5,4,3,2,1,0 FAILED

.....

If properly trained, perform the following procedure.

.....

The X-ray camera detected an invalid segment gain result during grayscale adjustment.

ID: 13
IP Address: 192.168.128.13
Port: 8080
Segment number: 0
Result: 1000.0
Low limit: 1.0
High limit: 5.0

Please do the following to recover:
1) Check for debris on the camera.

For a more detailed troubleshooting procedure, open the help system and navigate to: Service Documentation > Troubleshooting > X-ray Camera.

If problems persist, contact your authorized service representative.

Check for debris on camera array.

If problems persist, contact your authorized service representative.

OK

During CD&A / System Startup Error Message





Camera Grayscale Adjustment Troubleshooting

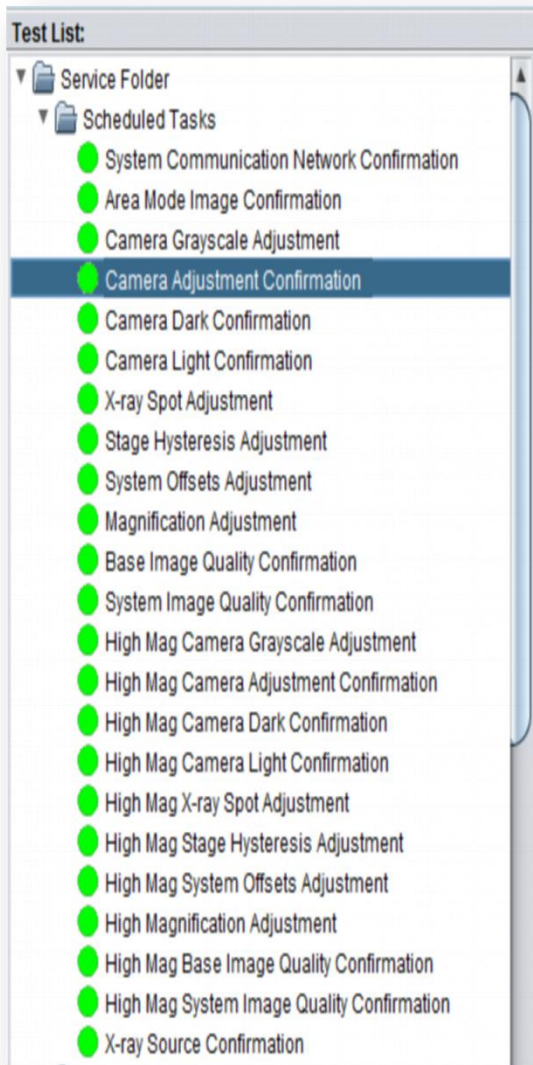
If one or two camera fail:

- Look for something in the image, such as valve tubing or debris on the camera
- Verify that the stage is in the correct position (center)
- Check failing devices area mode image
- Replace the failing device

If multiple or all camera fail:

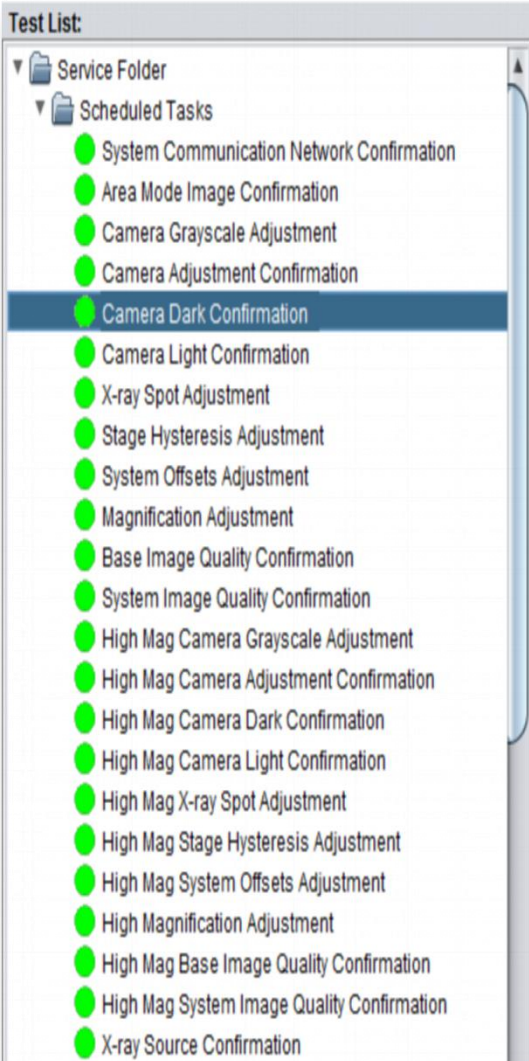
- Verify the inner barrier is opened
- Check for problems with the x-ray source (MFX tools, ensure no overload issue)

Camera Adjustment Confirmation



- Purpose** : Obtain the camera segment and pixel gain and offset values so that error could be detected earlier before hardware error occurs.
- : Maximum and minimum tolerated segment and pixel gain and offset values are stored in the limit file
- : After camera's have complete the segment and pixel gain and offset adjustment, it will compare with the customer limit file
- Method** : Check the four values from the Gray Scale Adjustment:
- 1.Segment gain & offset
 - 2.Pixel gain & offset
- : Identifies hardware that is operating close to its limits or is not adjusted
- Results** : Log result to the
- C:\Program Files\AXI System\v810\5.0\log\confirm\cameraAdjustment
- Remarks** : 13 text log will be generated once complete calibration

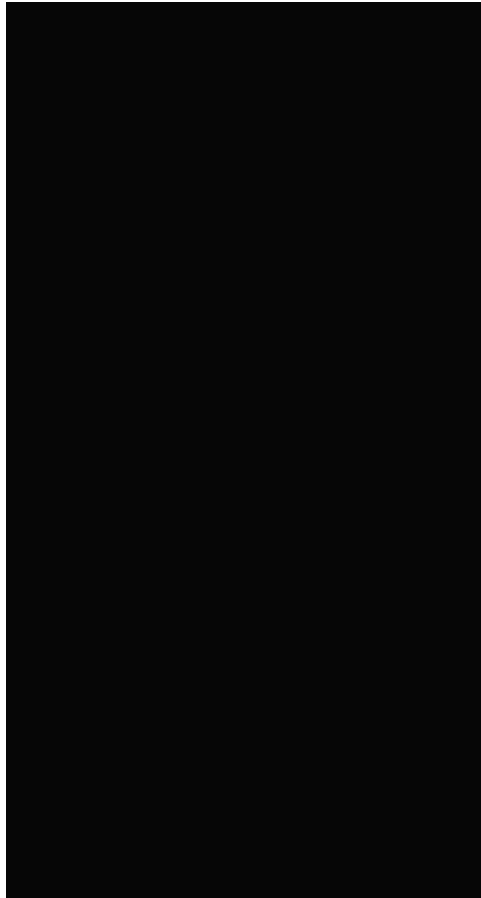
Camera Dark Confirmation



- Purpose**
- : Confirm the camera grayscale, dark value, works and the image are smooth in both X and Y
 - : Helps detect the noise in the system, X-Ray spot oscillation or failed grayscale calibrations
 - : Confirm that Camera Dark = 6 to 12 and +/- 2 gray levels
- Method**
- : Gather projection images using synthetic triggers with x-ray current on and inner barrier closed.
- Results**
- : Log result to the
C:\Program Files\AXI System\v810\5.0\log\confirm\cameraDark
- Remarks**
- : There is an image for each camera for forward and reverse
 - : 28 images and 1 text log will be generated for the camera dark



Camera Dark Confirmation Troubleshooting



Original Image



**Enhanced Image to
show defect**



Camera Dark Confirmation Troubleshooting

Note:

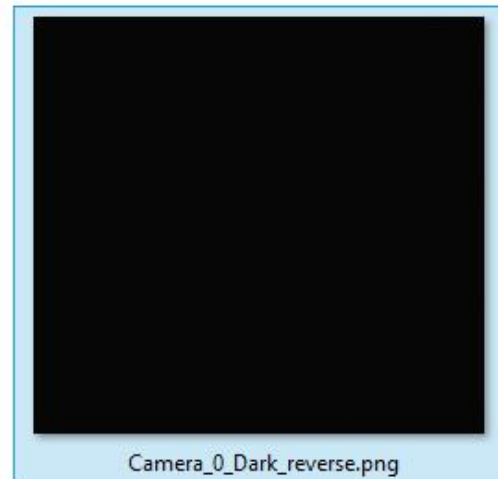
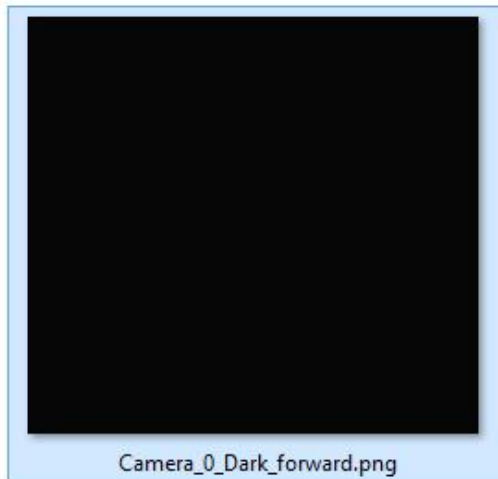
There is an image for each camera for forward and reverse

Each camera have 2 channels :

- a) Forward channels (used when stage is moving forward)
- b) Reverse channels (used when stage is moving reverse)

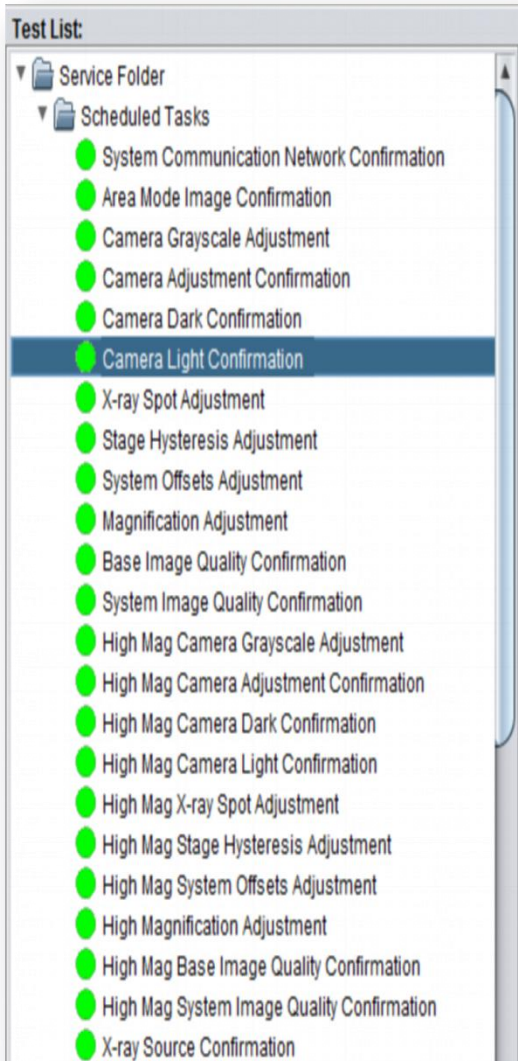
During this confirmation, the camera will use both channels with synthetic trigger but no motion of the stage.

There is no motion during Camera Dark Confirmation



- A light gray image could be caused by the inner barrier not closing
- Vertical line or streaking could be caused by noise on the camera cables
- Horizontal line or streaking could be caused by improper gray scale adjustment

Camera Light Confirmation



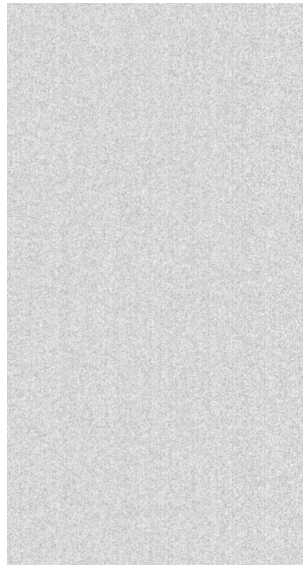
- Purpose** : Confirm the camera grayscale, light value, works and the image are smooth in both X and Y
- : Helps detect the noise in the system, X-Ray spot oscillation or failed grayscale calibrations
- : Grey level tested = 205 to 230
- Method** : Gather projection images using synthetic triggers with x-ray current on and inner barrier opened
- Results** : Log result to the
- C:\Program Files\AXI System\v810\5.0\log\confirm\cameraLight
- Remarks** : Stage position is important to expose anything that might have been masked by Gray Scale Adjustment
- : 28 images and 1 text log will be generated for the camera light



Camera Light Confirmation

Note:

- There is an image for each camera for forward and reverse
- All image files will stay White with noise considered as a good result
- Image with blank white (without noise) is the result of the calibration not updated
- Image files will be replaced and new text files will be generated every time the test is performed



Flat Image

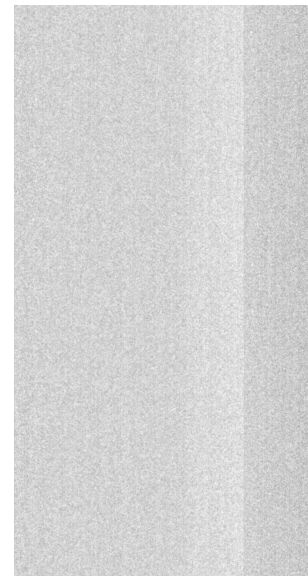
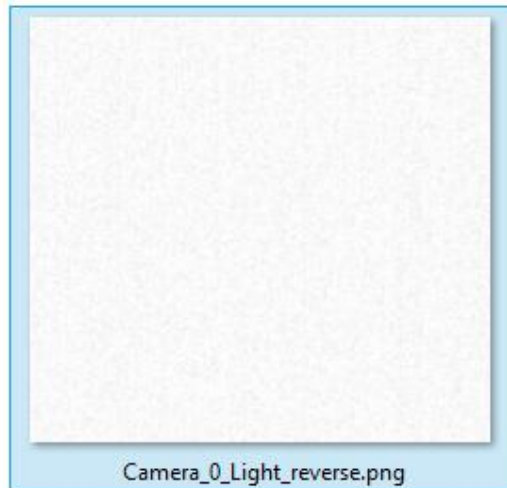


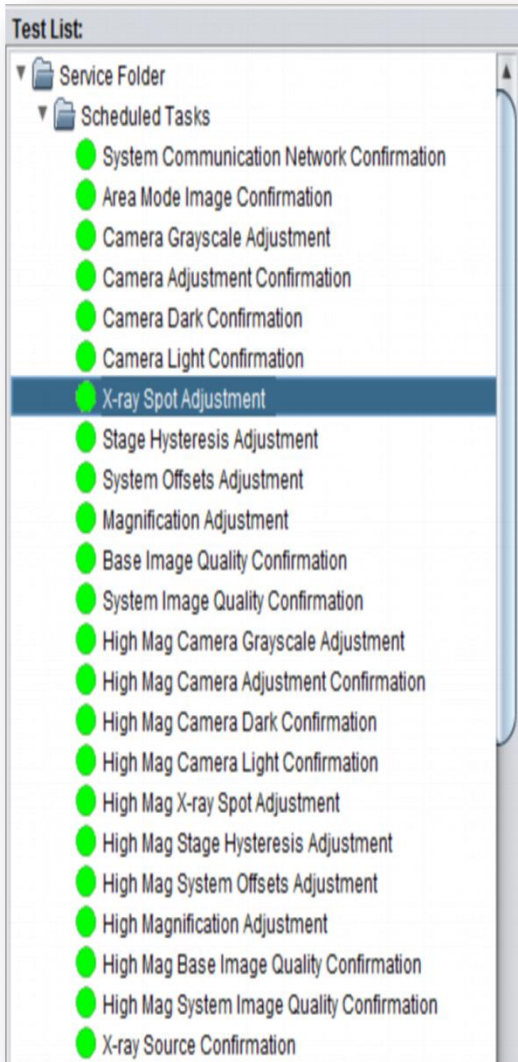
Image with
obvious defects

Camera Light Confirmation



- A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment
- Vertical line or streaking could be caused by noise on the camera cables
- Horizontal line or streaking could be caused by improper gray scale adjustment, debris on the camera array, or improper stage position
- If multiple camera light confirmation fail, need to reboot the trigger board power

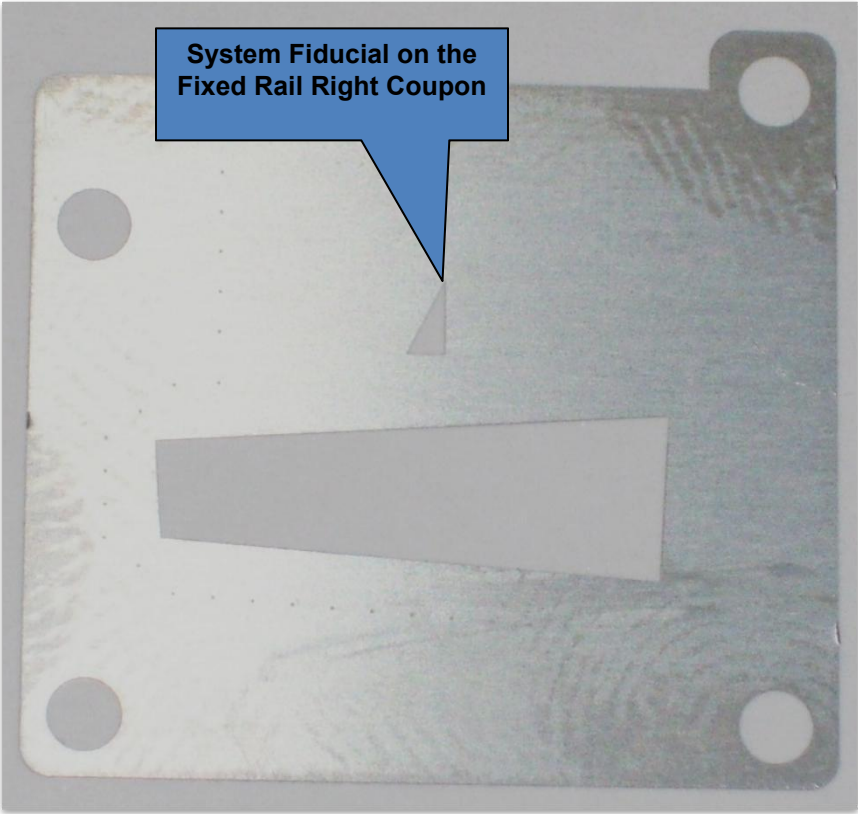
X-Ray Spot Adjustment



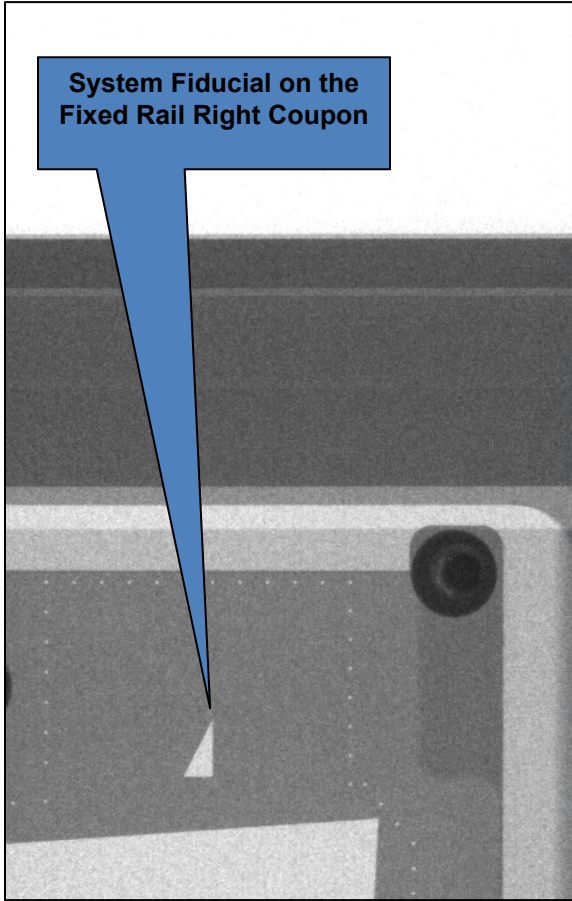
- Purpose**
- : Calculates the X/Y stage position in relationship to the X-Ray spot via the coupon over the camera 0
 - : Determines the required X/Y stage offset to ensure proper stage positioning
 - : Calculates the XY position of the stage with the Fixed Rail Right Fiducial Coupon centered over the camera array
- Method**
- : Uses motion triggers and searches the image for the system fiducial
 - : Calculate the XY offset from expected to actual based on the hardware.calib
- Results**
- : Log result to the
C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot
- Remarks**
- : Records the values in the hardware.calib
 - : 2 image files and 2 test file will be generated (each for difference Mag) :
cam_0.png and cam_high_0.png
 - : 2 original template files
 - : S2EX X-ray spot can only be optimized using the X-ray spot alignment jig
 - : X-ray spot will be different for each machines



X-Ray Spot Location Adjustment



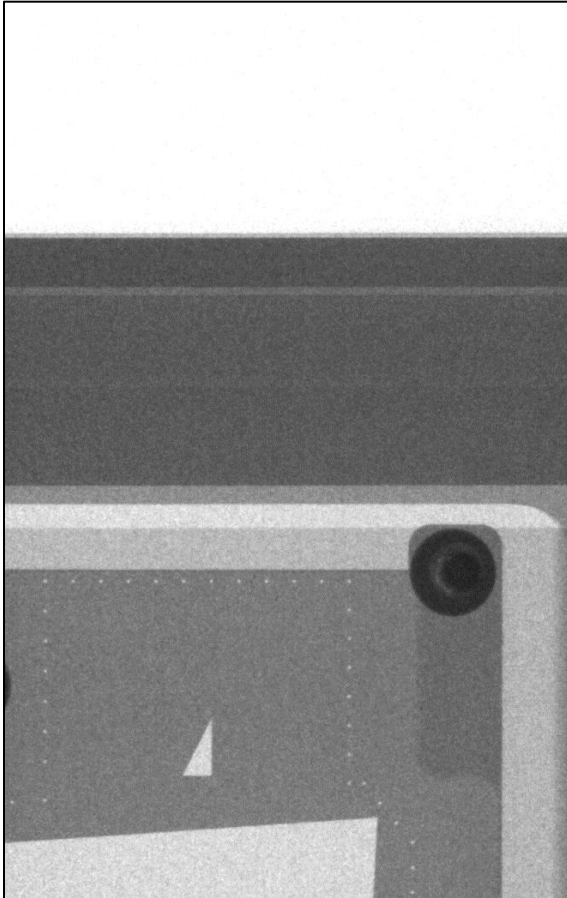
Fiducial coupon



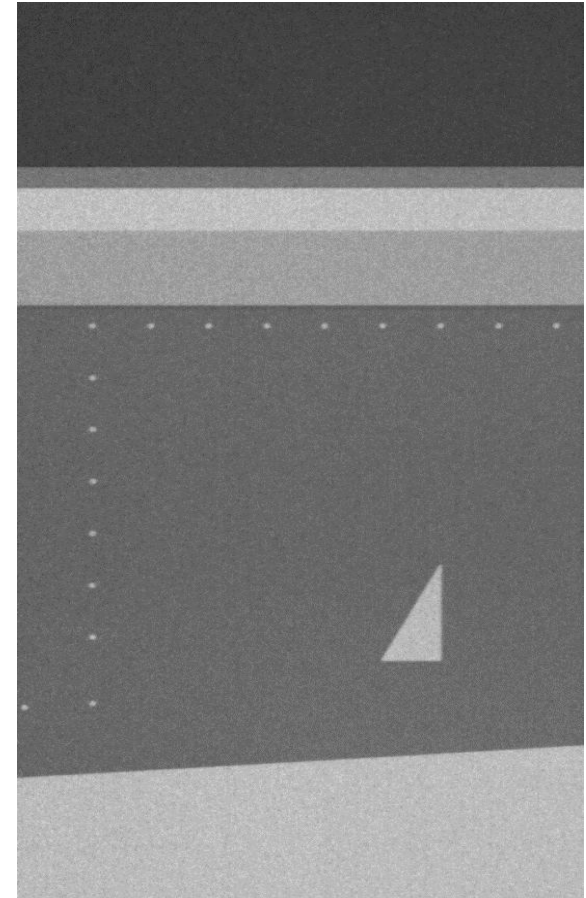
X-Ray spot



X-Ray Spot Adjustment (23μ and 11μ)



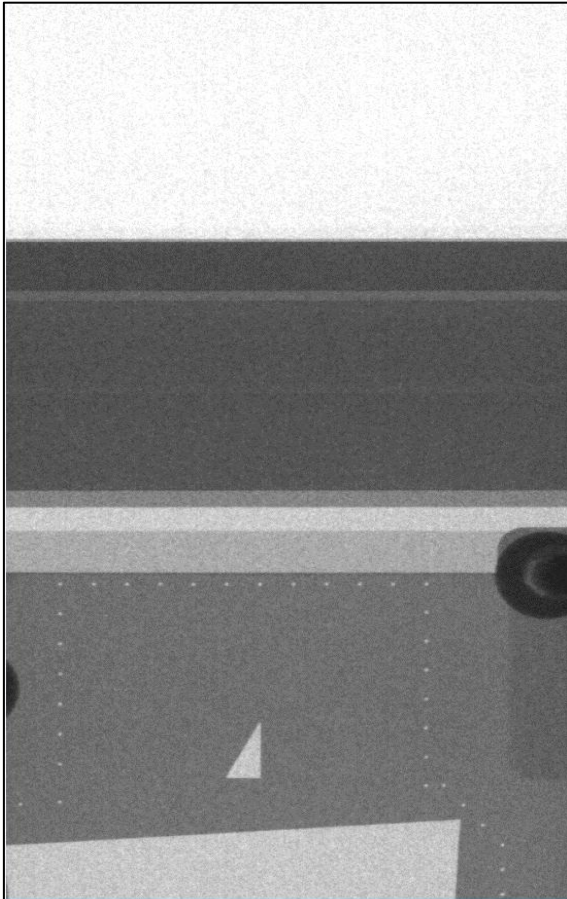
23μm



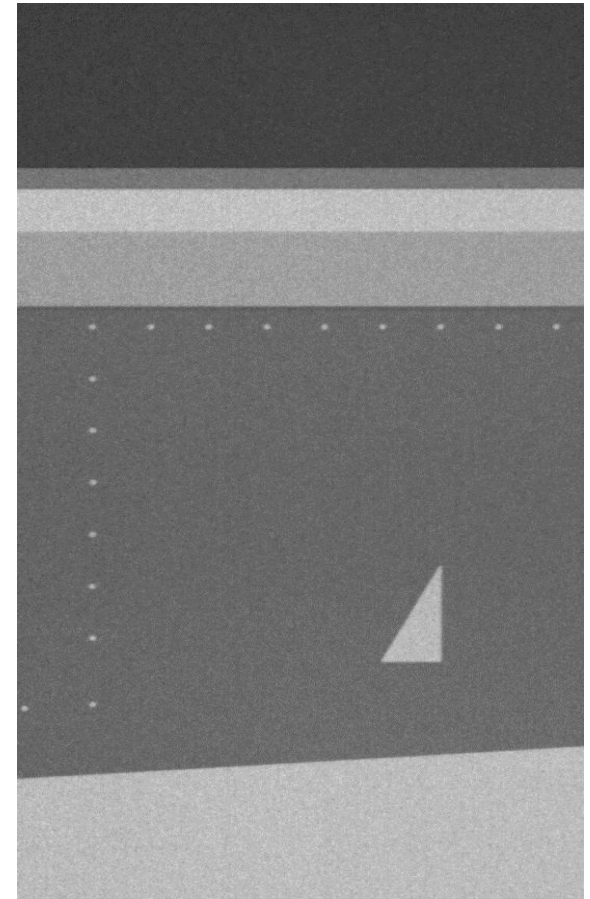
11μm



X-Ray Spot Adjustment (19μ and 11μm)



19μm



11μm

X-Ray Spot Adjustment Troubleshooting

- A black or dark images could be caused by the inner barrier not opening, defective X-Ray source or defective Camera 0
- Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib, or motion trigger problem
- Vertical line or streaking could be caused by improper greyscale adjustment or debris on the camera array
- Horizontal line or streaking could be caused by a motion trigger problem or other image capture issues.



X-Ray Spot Adjustment Troubleshooting

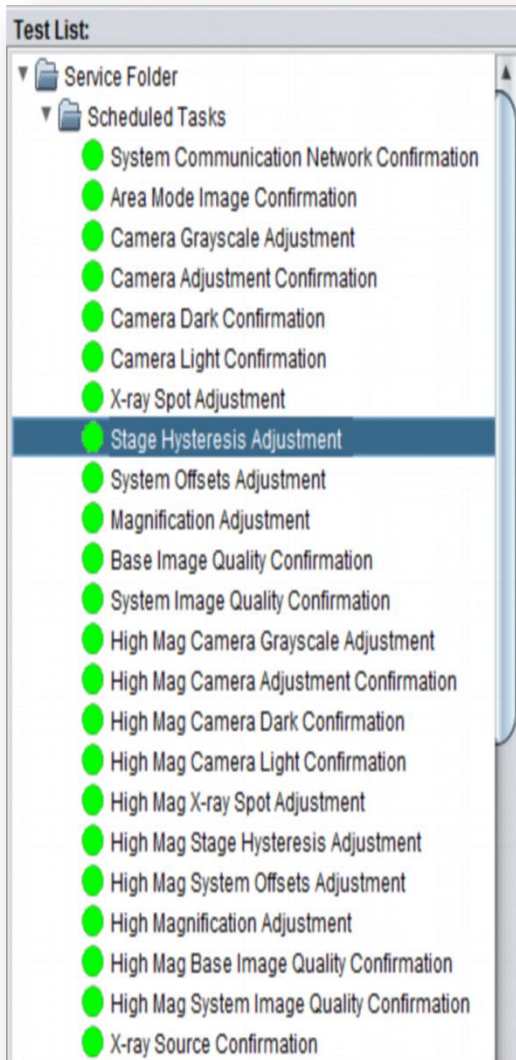
Error Message:

1. Stage Hysteresis Adjustment failed
2. System Offset Adjustment failed
3. Base Image Confirmation intermittently fail during CDnA
4. Insufficient trigger in random camera's during CDnA or production
5. Stroke not enough during CDnA

Root Cause: X-Ray spot is not optimized

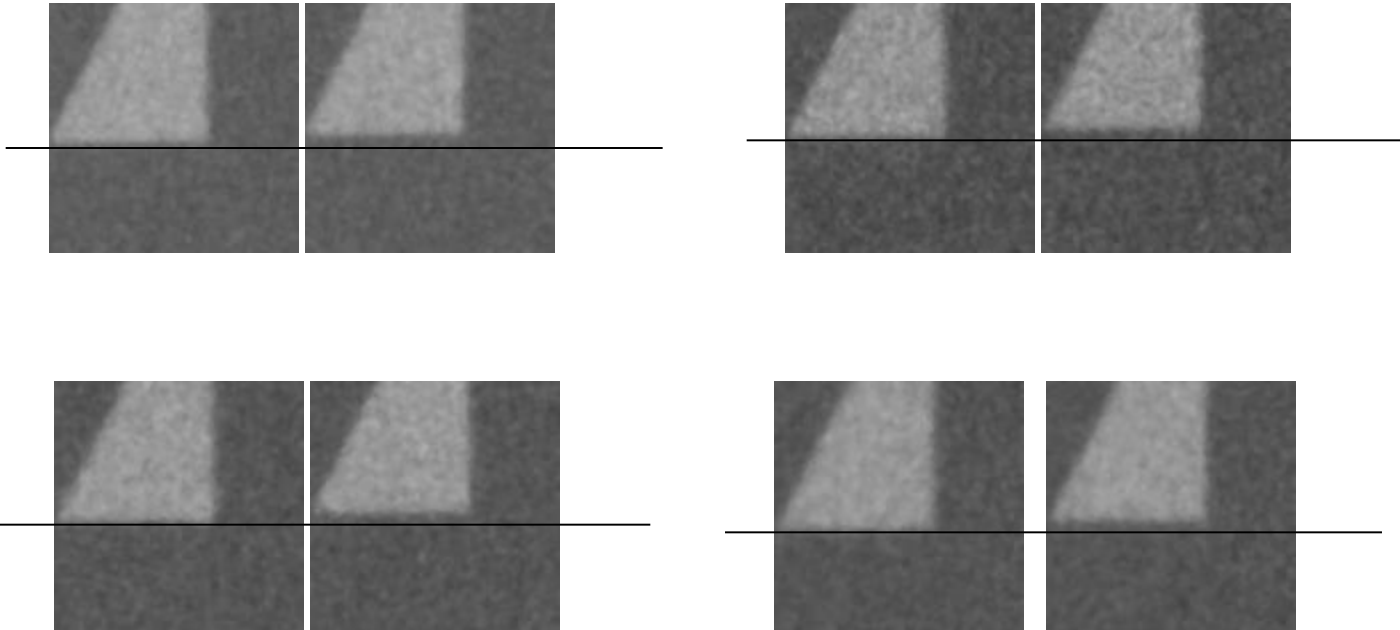
Solution: Optimize the X-ray spot to the correct position for both magnification using the x-ray spot jig for (23 μ m&11 μ m /19 μ m&11 μ m)

Stage Hysteresis Adjustment



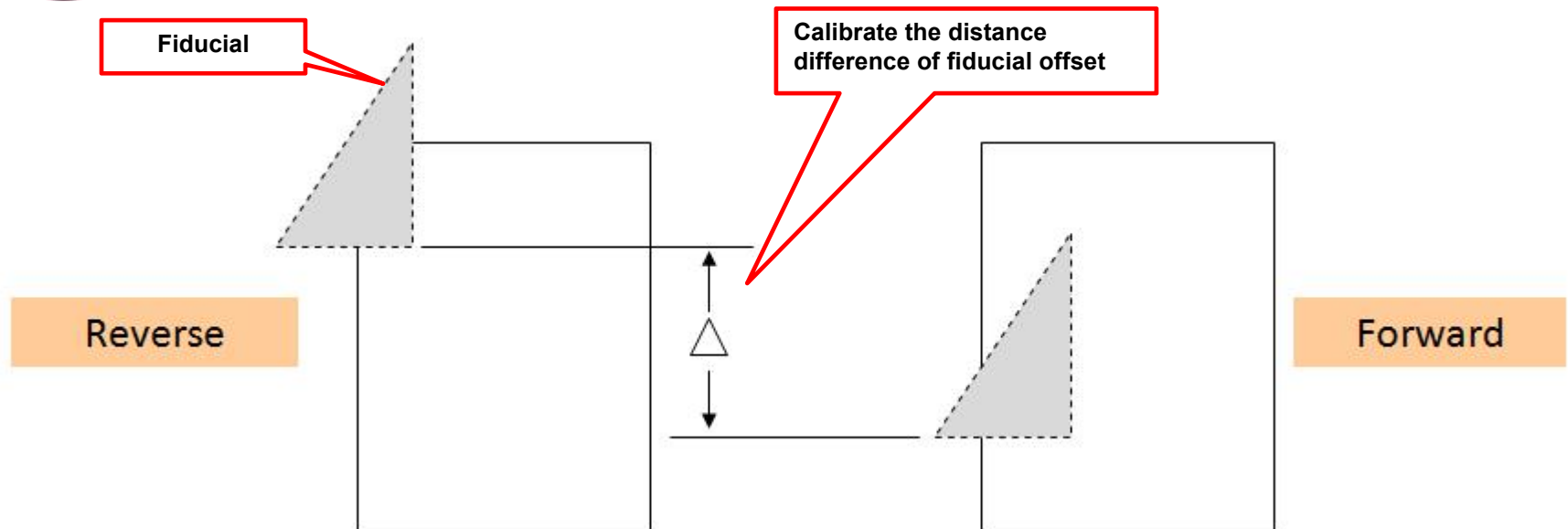
- Purpose** : Adjusts for differences in Y-Axis motion between forward and reverse for image reconstruction
- : Minimize the offset gap for Y-axis movement front and reverse action
- Method** : 2 images being captured from 14 camera's when stage moving forward and backward for each camera. Then generate the offset by comparing both images
- : Starting stage position is the same as X-Ray Spot Location Adjustment
- : Generates 2 projection images of the system fiducial for each camera (one forward and one reverse)
- : Use the Fixed Rail Right Fiducial Coupon for image capturing
- Results** : Log result to the
C:\Program Files\AXI System\v810\5.0\log\calib\StageHysteresis
- Remarks** : Calculates the averages of the 14 values and records it in hardware.calib file:
yAxisHysteresisOffsetInNanometers
- : The offset different must be within the 0 to 800000 nanometers

Stage Hysteresis Adjustment



04/22/2015 6:42:50 PM
High Mag Stage Hysteresis
Adjustment,Hysteresis,
,,pixels,9.231646,,nm,101548,

Stage Hysteresis Adjustment



- Each camera will generate 2 image file itch cam_XX_up and cam_XX_down once the test is complete test

Error Message : Stage Hysteresis Failed

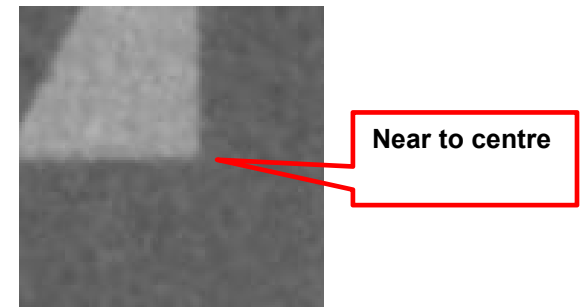
Possible root cause : Z-Height tube is not at optimized height (too high or too low)

: Camera device failed

Solution : Adjust back the tube to correct height

- Correct tube height will be projected image as shown.

- The Fiducial will allocated at the center of the image



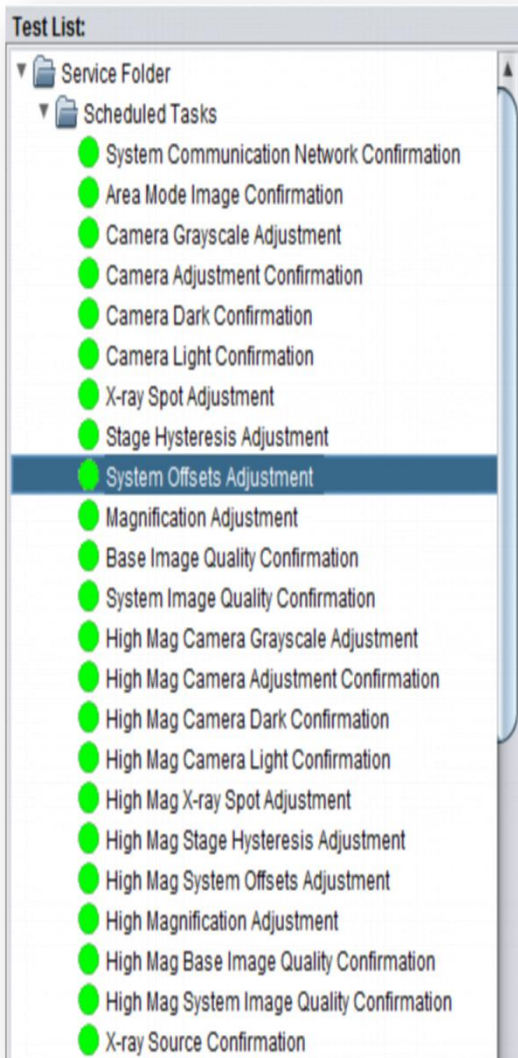


Stage Hysteresis Adjustment Troubleshooting

Open the image file: C:\Program Files\AXI System\v810\5.0\log\calib\StageHysteresis

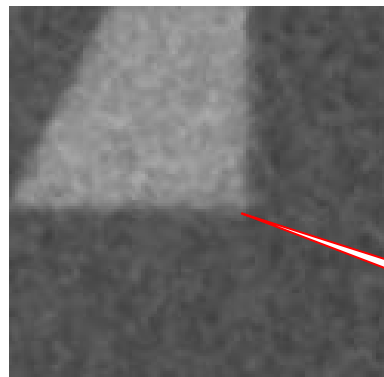
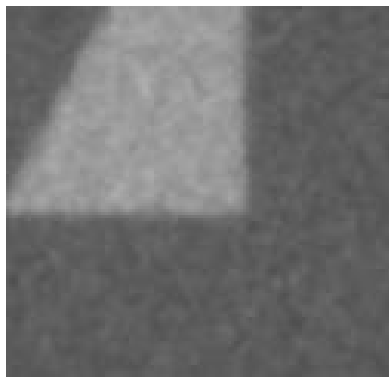
- 1. A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment**
- 2. Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib**
- 3. Vertical line or streaking could be caused by improper grayscale adjustment or other image capturing issues**
- 4. Y-Axis mechanical problem**
- 5. Dismantle the X-ray tube and re-install the X-Ray tube correctly**

System Offset Adjustment

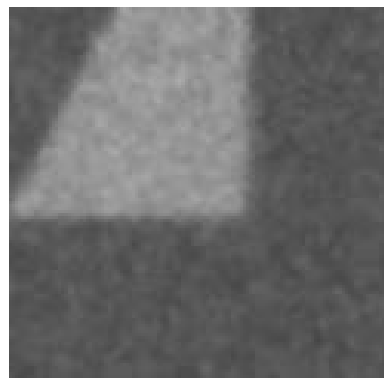
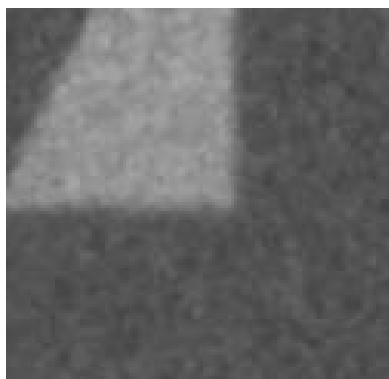


- Purpose** : Determines the X/Y stage positions from the center camera to each of the other cameras for image reconstruction
- : The value is only used during reconstruction. It is passed to the IRP's and is used to do a final image alignment during the reconstruction process
- Method** : Moves the Fixed Rail Right Fiducial over each camera to determine four values:
- X&Y stage
 - Camera Column & Row
- Results** : Log result to the
- C:\Program Files\AXI System\v810\5.0\log\calib\SystemOffsets
- Remarks** : 14 images and 1 text log will be generated

System Offset Adjustment



Near to the
center



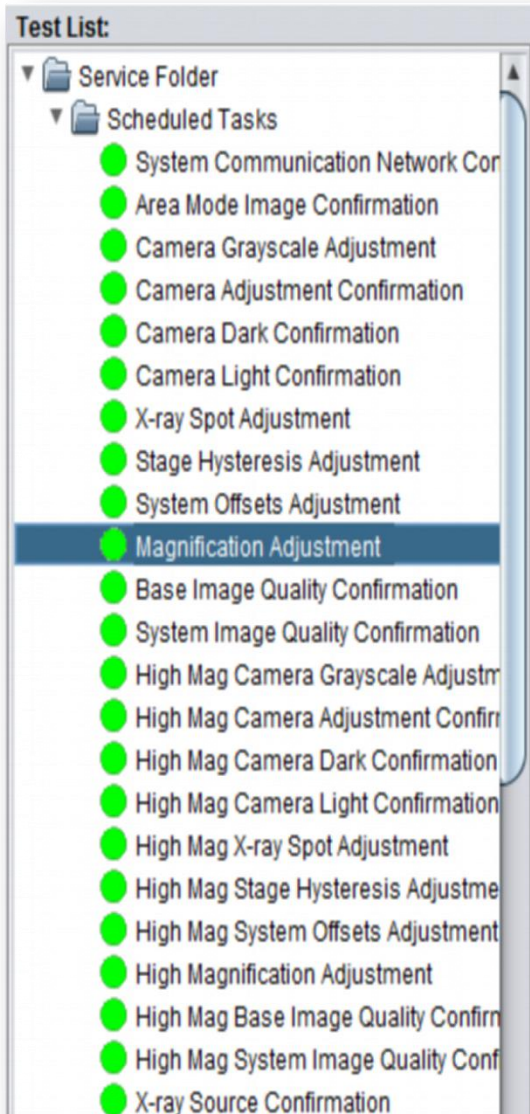


System Hysteresis Adjustment Troubleshooting

Open the image file: C:\Program Files\AXI System\v810\5.0\log\calib\StageHysteresis

- 1. A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment**
- 2. Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib**
- 3. Vertical line or streaking could be caused by improper grayscale adjustment or other image capturing issues**
- 4. Horizontal line or streaking could be caused by a motion trigger problem or other image capture issues.**

Magnification Adjustment



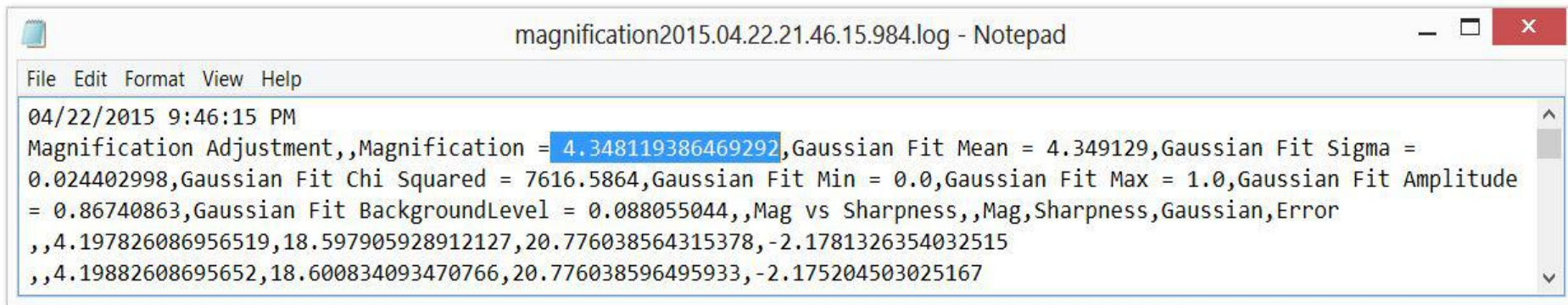
- Purpose** : Determines the Reference Plane magnification.
- : Generates reconstructed images of the vertical edge on the Fixed Rail Right Fiducial triangle using different values for Magnification and determines which value generates sharpest images
- Results** : Log result to the C:\Program Files\AXI System\v810\5.0\log\calib\XraySpot



Magnification Adjustment (23 μ m & 11 μ m)

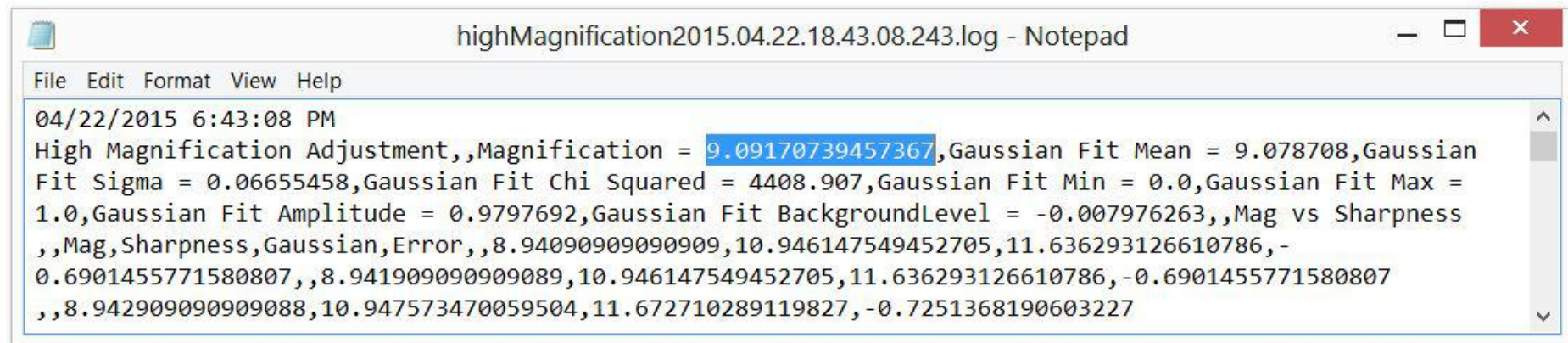
Open the latest test.log and check the actual magnification

Log Mag: magnificationYYYY.MM.DD.HH.MM.SS.XXX.log



```
magnification2015.04.22.21.46.15.984.log - Notepad
File Edit Format View Help
04/22/2015 9:46:15 PM
Magnification Adjustment,,Magnification = 4.348119386469292,Gaussian Fit Mean = 4.349129,Gaussian Fit Sigma =
0.024402998,Gaussian Fit Chi Squared = 7616.5864,Gaussian Fit Min = 0.0,Gaussian Fit Max = 1.0,Gaussian Fit Amplitude
= 0.86740863,Gaussian Fit BackgroundLevel = 0.08805504,,Mag vs Sharpness,,Mag,Sharpness,Gaussian>Error
,,4.197826086956519,18.597905928912127,20.776038564315378,-2.1781326354032515
,,4.19882608695652,18.600834093470766,20.776038596495933,-2.175204503025167
```

High Mag: highmagnificationYYYY.MM.DD.HH.MM.SS.XXX.log



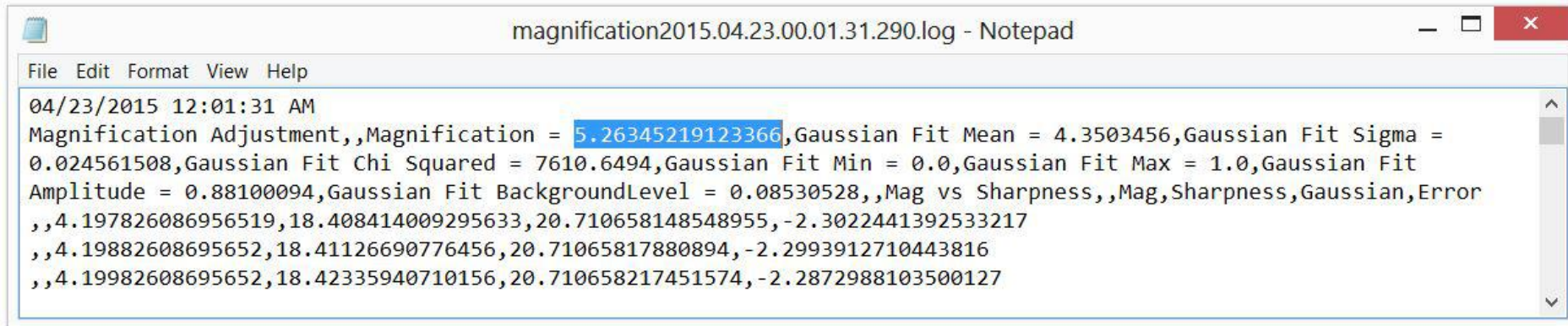
```
highMagnification2015.04.22.18.43.08.243.log - Notepad
File Edit Format View Help
04/22/2015 6:43:08 PM
High Magnification Adjustment,,Magnification = 9.09170739457367,Gaussian Fit Mean = 9.078708,Gaussian
Fit Sigma = 0.06655458,Gaussian Fit Chi Squared = 4408.907,Gaussian Fit Min = 0.0,Gaussian Fit Max =
1.0,Gaussian Fit Amplitude = 0.9797692,Gaussian Fit BackgroundLevel = -0.007976263,,Mag vs Sharpness
,,Mag,Sharpness,Gaussian>Error,,8.94090909090909,10.946147549452705,11.636293126610786,-
0.6901455771580807,,8.941909090909089,10.946147549452705,11.636293126610786,-0.6901455771580807
,,8.942909090909088,10.947573470059504,11.672710289119827,-0.7251368190603227
```



Magnification Adjustment (19 μ m & 11 μ m)

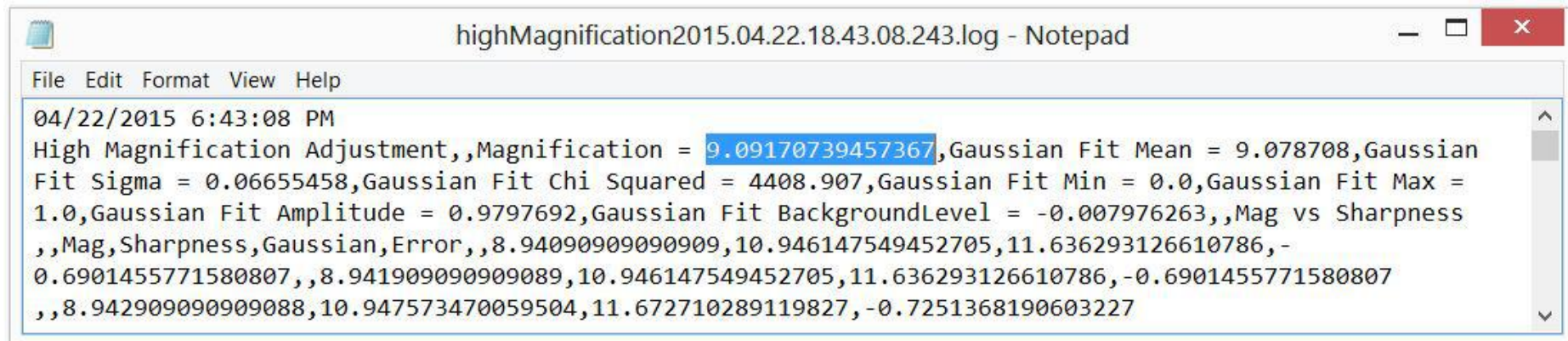
Open the latest test.log and check the actual magnification

Log Mag: magnificationYYYY.MM.DD.HH.MM.SS.XXX.log



```
magnification2015.04.23.00.01.31.290.log - Notepad
File Edit Format View Help
04/23/2015 12:01:31 AM
Magnification Adjustment,,Magnification = 5.26345219123366,Gaussian Fit Mean = 4.3503456,Gaussian Fit Sigma =
0.024561508,Gaussian Fit Chi Squared = 7610.6494,Gaussian Fit Min = 0.0,Gaussian Fit Max = 1.0,Gaussian Fit
Amplitude = 0.88100094,Gaussian Fit BackgroundLevel = 0.08530528,,Mag vs Sharpness,,Mag,Sharpness,Gaussian>Error
,,4.197826086956519,18.408414009295633,20.710658148548955,-2.3022441392533217
,,4.19882608695652,18.41126690776456,20.71065817880894,-2.2993912710443816
,,4.19982608695652,18.42335940710156,20.710658217451574,-2.2872988103500127
```

High Mag: highmagnificationYYYY.MM.DD.HH.MM.SS.XXX.log



```
highMagnification2015.04.22.18.43.08.243.log - Notepad
File Edit Format View Help
04/22/2015 6:43:08 PM
High Magnification Adjustment,,Magnification = 9.09170739457367,Gaussian Fit Mean = 9.078708,Gaussian
Fit Sigma = 0.06655458,Gaussian Fit Chi Squared = 4408.907,Gaussian Fit Min = 0.0,Gaussian Fit Max =
1.0,Gaussian Fit Amplitude = 0.9797692,Gaussian Fit BackgroundLevel = -0.007976263,,Mag vs Sharpness
,,Mag,Sharpness,Gaussian>Error,,8.94090909090909,10.946147549452705,11.636293126610786,-
0.6901455771580807,,8.94190909090909,10.946147549452705,11.636293126610786,-0.6901455771580807
,,8.942909090909088,10.947573470059504,11.672710289119827,-0.7251368190603227
```



Magnification Adjustment- Example Image



Mag at 4.31
Clearly out of
focus

Best Mag at 4.34
(in this example)
Sharpest line

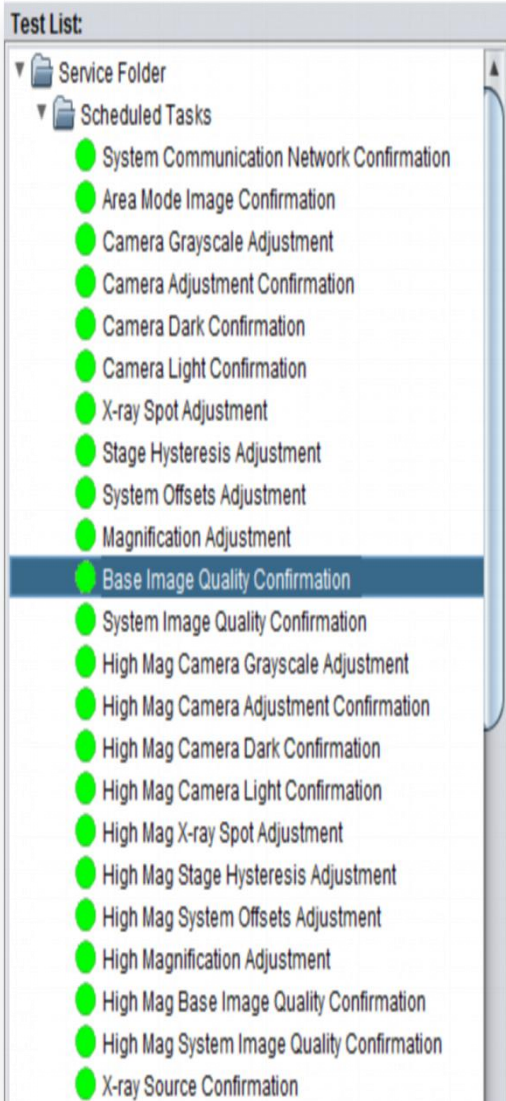
Mag at 4.39
Again, out of focus



Magnification Adjustment Troubleshooting

- **A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment**
- **Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib file, a motion or trigger problem**
- **Vertical line or streaking could be caused by improper greyscale adjustment or debris on the camera array**
- **Horizontal line or streaking could be caused by a motion trigger problem or other image capture issues**

Base Image Quality Confirmation



Purpose : Determines Image Quality by generating projection images of the quadrilateral on the Fixed Rail Right Fiducial coupon.

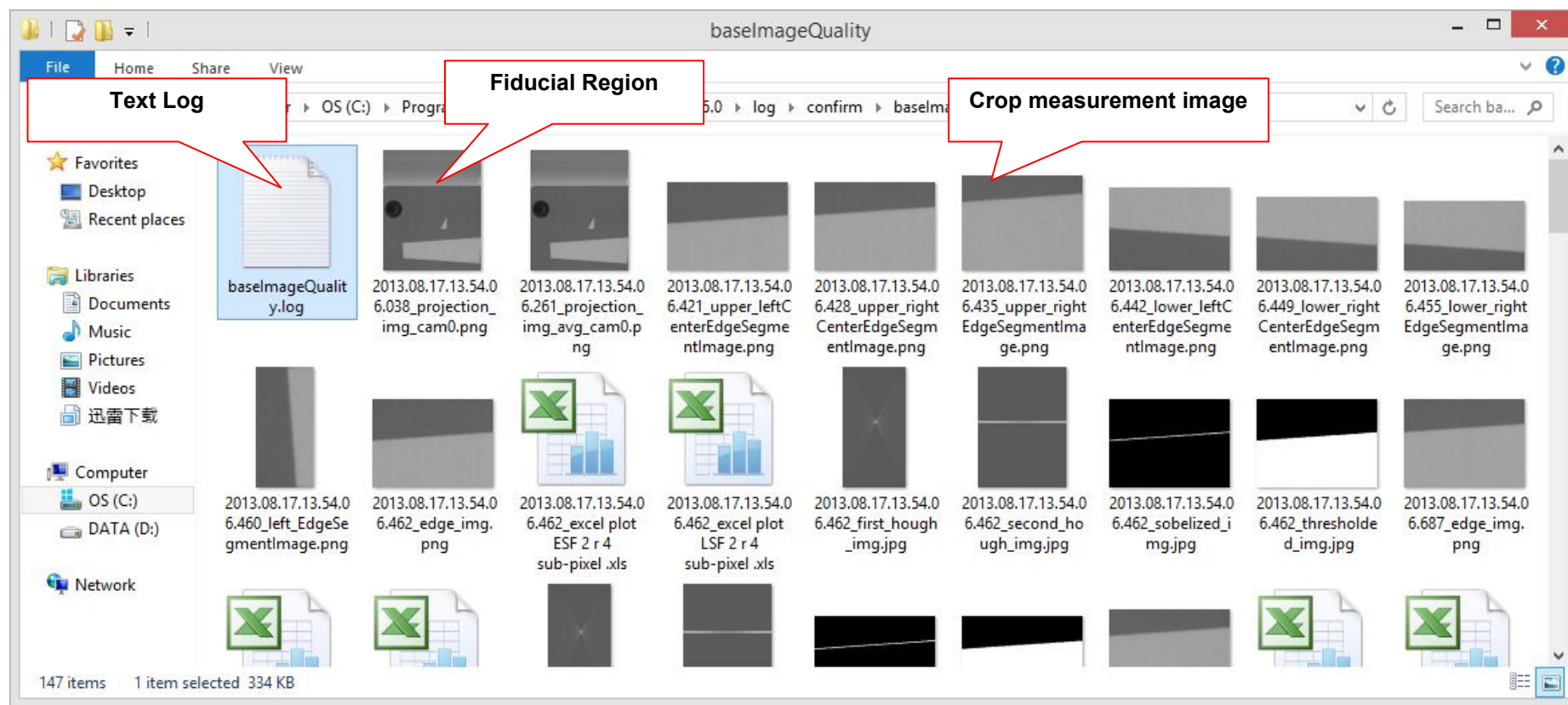
Method : Average 16 images over the center camera (Only camera 0)
: Measure the sharpness of the horizontal edges and one vertical edge of the quadrilateral
: Compare the values with the limits to determine pass/ fail

Results : Log result to the
: C:\Program Files\AXI System\v810\5.0\log\confirm\baseImageQuality
: C:\Program Files\AXI System\v810\5.0\log\confirm\baseImageQualityForHighMag

Base Image Quality Confirmation

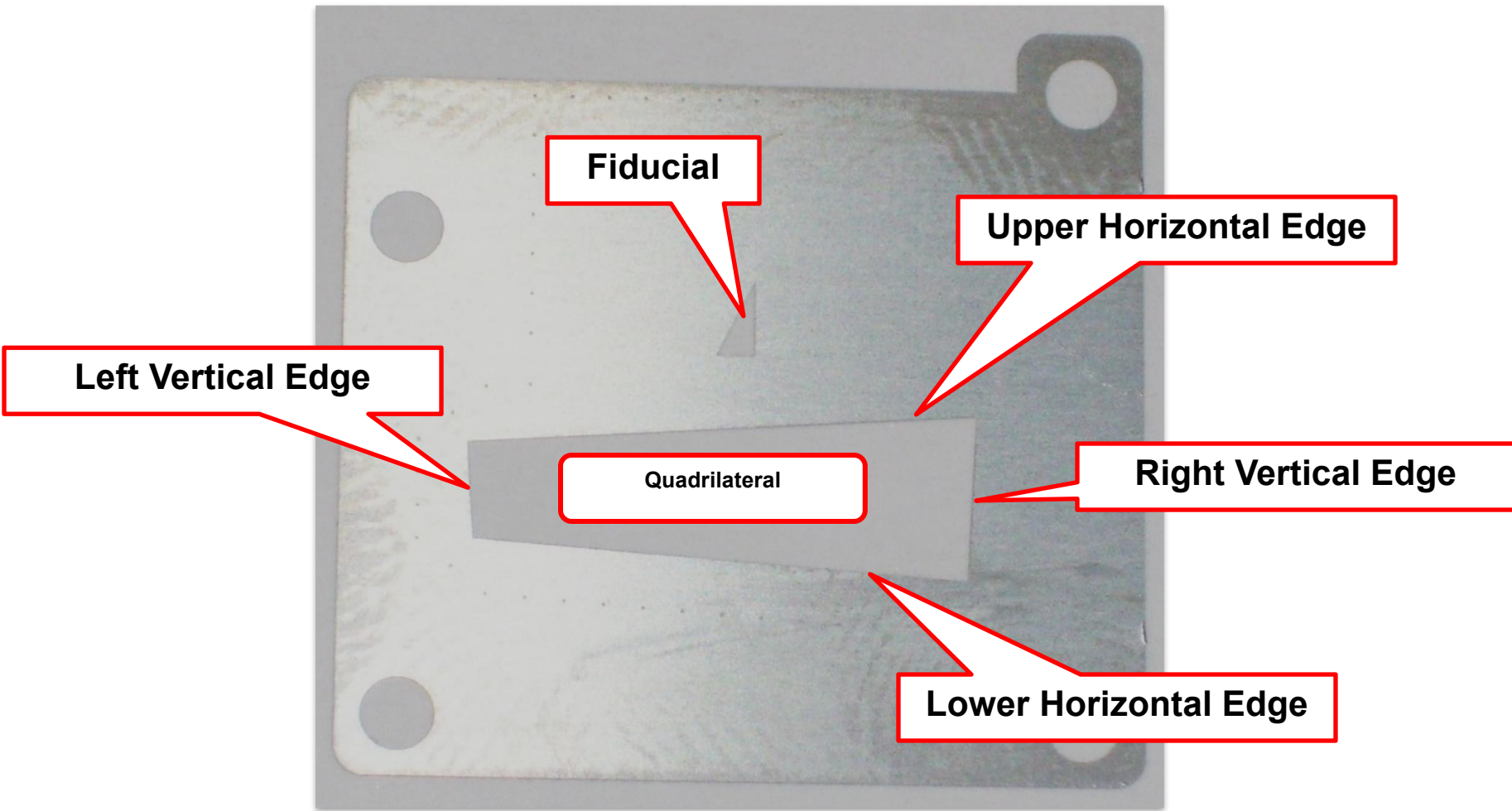
If this confirmation fails, it logs the last failing image

For troubleshooting, turn on logging in the software.config file (saveAllConfirmationImages=true) so it will log all images



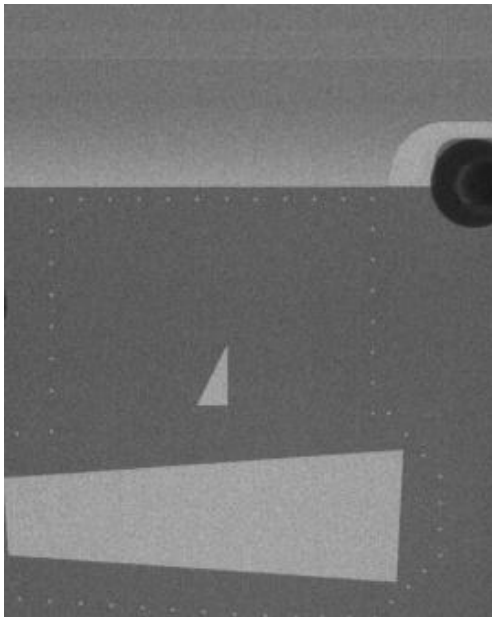


Base Image Quality Confirmation – Fiducial Coupon

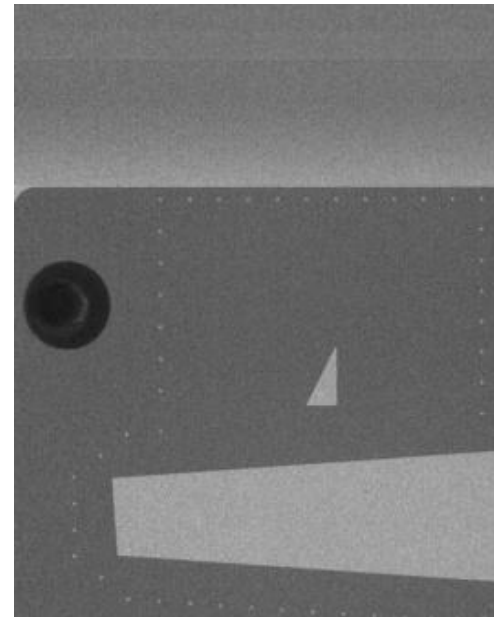




Base Image Quality Confirmation – Example Image



Region A



Region B

During test, system will capture 2 fiducial region projection image and crop 7 image at each region and measure sharpness value and compare with the limit spec.

Spec for Low Mag (23 μ m & 19 μ m) :

The vertical edge and horizontal edge value should fall between **2.0 – 6.0**

Spec for High Mag (11 μ m) :

The vertical edge and horizontal edge value should fall between **2.0 – 7.5**



Base Image Quality Confirmation

➤ Scroll to latest update date and check on the 3rd column on each crop image result

Date & Time

Crop area

Sharpness
Result

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
247	1/10/2015	4:32:41 PM	top	line	far	left	for	camera	0	93.81	4.692	3.29	top	line	left	of	center	for	camera	0	94.05	4.464	3.302	
248	1/11/2015	2:57:54 PM	top	line	far	left	for	camera	0	93.81	4.529	3.177	top	line	left	of	center	for	camera	0	94.05	4.404	3.084	
249	1/12/2015	2:58:41 PM	top	line	far	left	for	camera	0	93.81	4.451	3.142	top	line	left	of	center	for	camera	0	94.05	4.262	3.201	
250	1/13/2015	2:59:20 PM	top	line	far	left	for	camera	0	93.81	4.415	3.162	top	line	left	of	center	for	camera	0	94.08	4.454	3.186	
251	1/13/2015	3:07:45 PM	top	line	far	left	for	camera	0	93.81	4.628	3.118	top	line	left	of	center	for	camera	0	94.05	4.429	3.105	
252	1/14/2015	7:44:49 PM	top	line	far	left	for	camera	0	93.82	4.508	3.397	top	line	left	of	center	for	camera	0	94.05	4.48	3.342	
253	1/15/2015	2:44:30 PM	top	line	far	left	for	camera	0	93.93	4.572	3.117	top	line	left	of	center	for	camera	0	94.05	4.601	3.117	
254	1/16/2015	6:48:15 AM	top	line	far	left	for	camera	0	93.81	4.561	3.328	top	line	left	of	center	for	camera	0	94.05	4.598	3.267	
255	1/16/2015	6:49:50 AM	top	line	far	left	for	camera	0	93.93	4.381	3.221	top	line	left	of	center	for	camera	0	94.05	4.48	3.176	
256	1/16/2015	6:51:23 AM	top	line	far	left	for	camera	0	93.88	4.492	3.293	top	line	left	of	center	for	camera	0	94.05	4.383	3.242	
257				1										2										

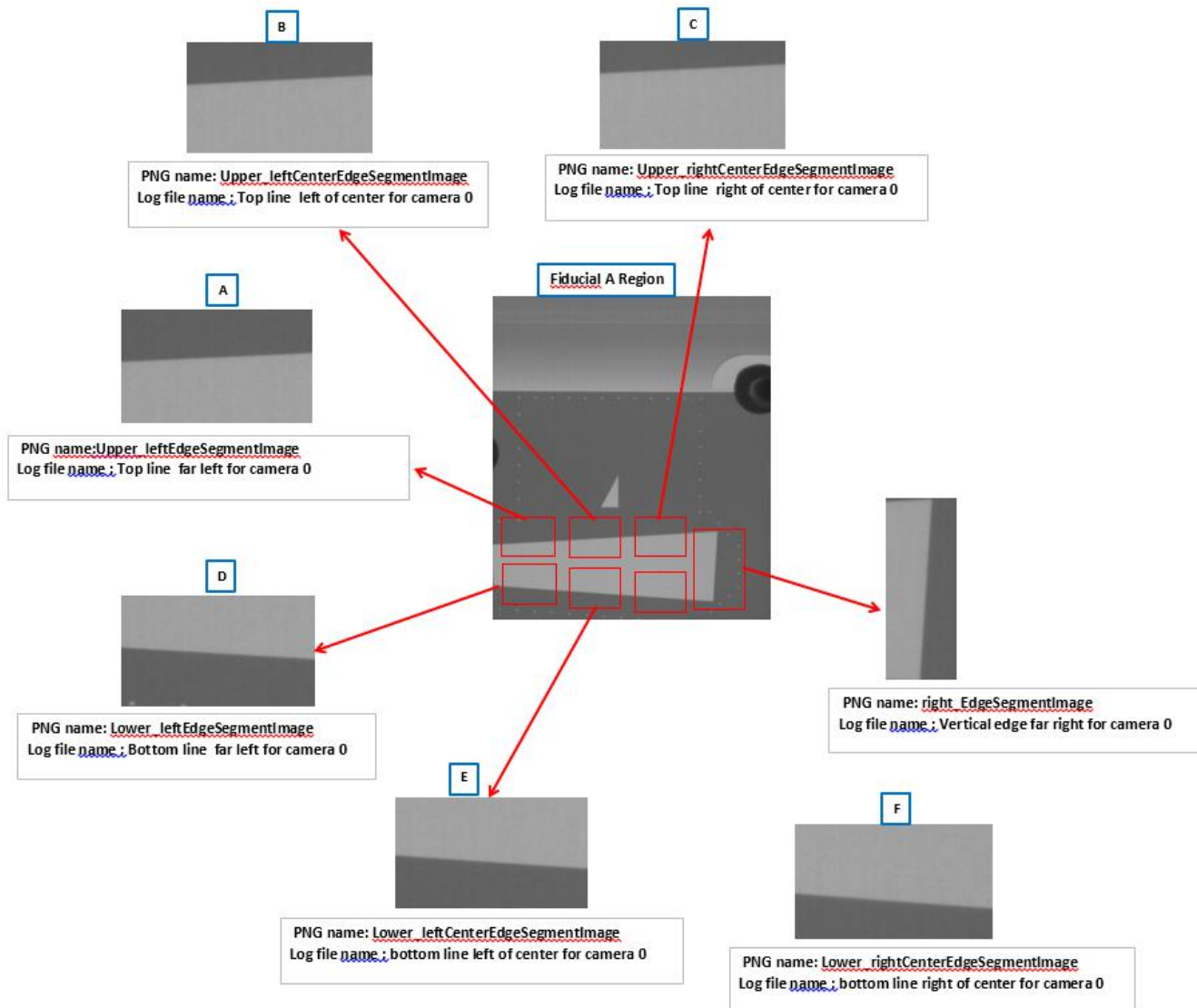
Camera ID

➤ If measurement failed, log files will show blank

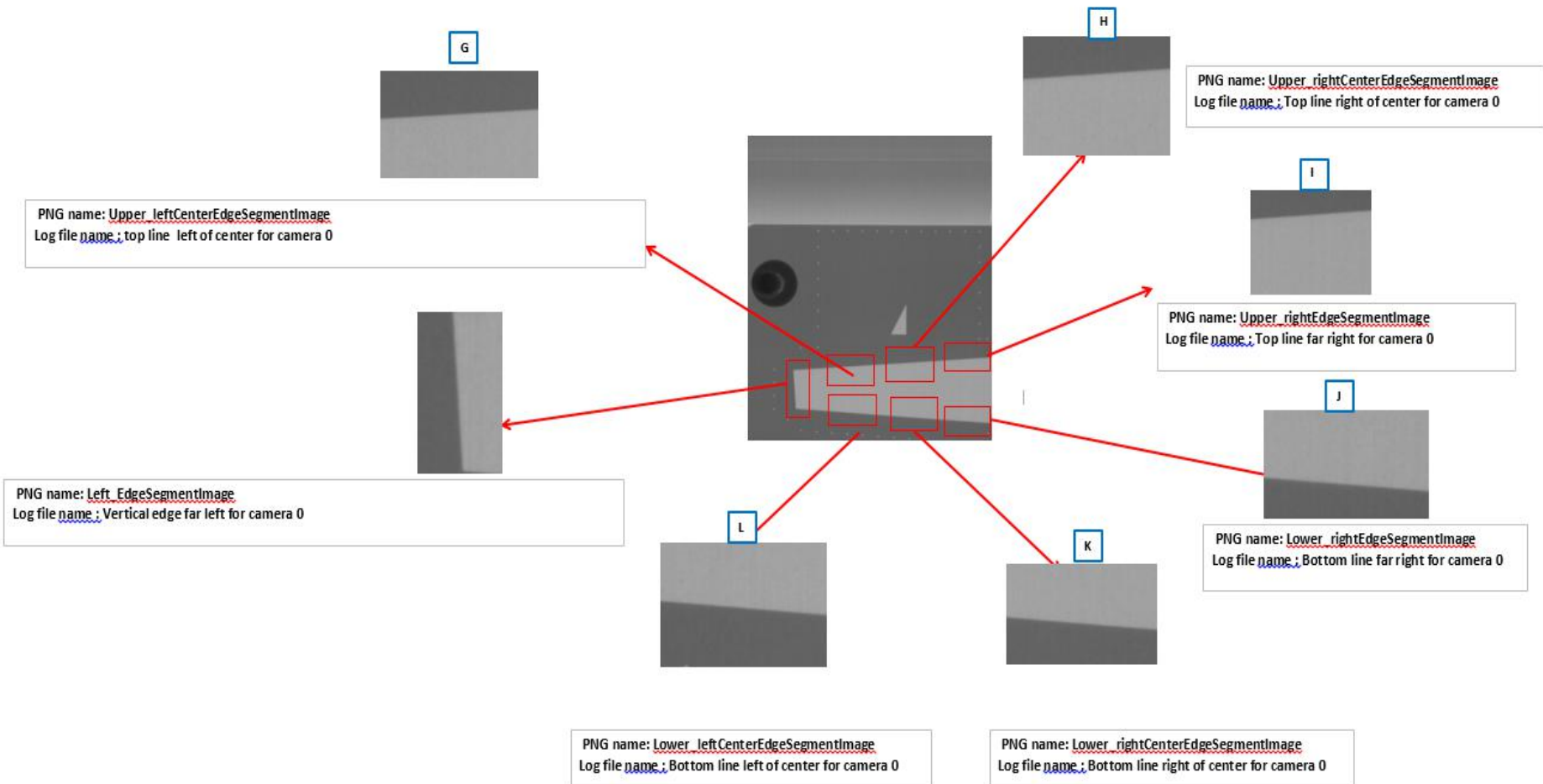
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
1	5/6/2014	10:07:16 AM	top	line	far	left	for	camera	0	93.88	4.269	3.321	top	line	left	of	center	for	camera	0	94.05	4.229	3.363	
2	7/18/2014	9:57:16 AM																						
3	7/18/2014	10:04:36 AM	top	line	far	left	for	camera	0	93.97	4.125	3.188	top	line	left	of	center	for	camera	0	94.05	4.198	3.234	
4	7/18/2014	10:10:11 AM	top	line	far	left	for	camera	0	93.95	4.216	3.09	top	line	left	of	center	for	camera	0	93.9	4.307	3.201	

Blank

Base Image Quality Confirmation



Base Image Quality Confirmation

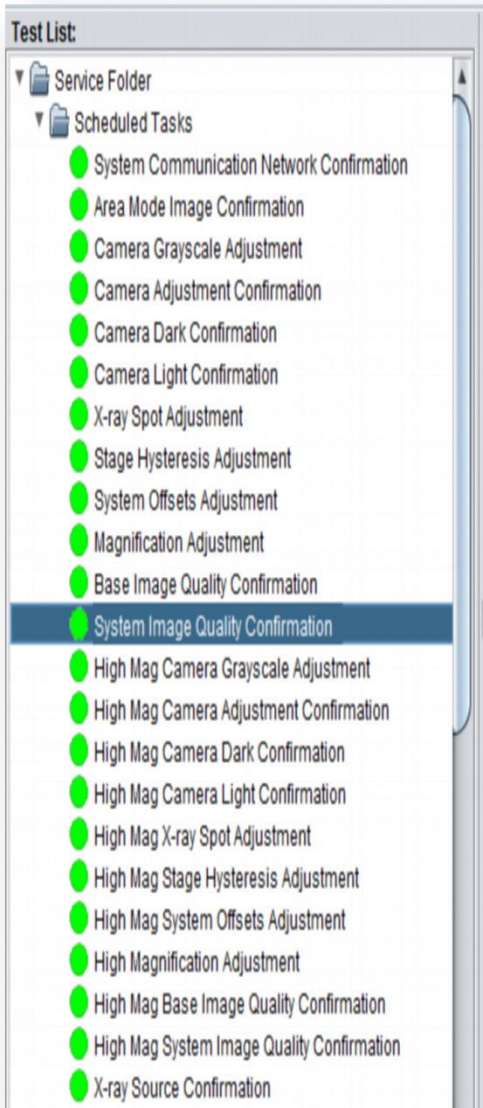




Base Image Quality Confirmation Troubleshooting

- Check on the baseImageQuality.log
- A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment
- Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib, a motion or trigger problem
- Vertical line or streaking could be caused by improper greyscale adjustment or debris on the camera array
- Horizontal line or streaking could be caused by a motion trigger problem or other image capture issues
- Checking X-ray Spot and perform X-Ray tube alignment if necessary

System Image Quality Confirmation



- Purpose** : Determines system level image quality
- : Most comprehensive Image Quality check which involves all subsystems
- Method** : Generates reconstructed images of the Quadrilateral on the **Fixed Rail Left Fiducial Coupon**. This coupon is at a different Z height than the Fixed Rail Right Fiducial Coupon.
- : Averages 3 images of the horizontal edge and one image of the vertical edge then compares the sharpness result with limits to determine pass/fail
- : Use 14 camera to snap image and reconstruct.
- Results** : Log result to the
- : C:\Program Files\AXI System\v810\5.0\log\confirm\systemImageQuality
- : C:\Program Files\AXI System\v810\5.0\log\confirm\SystemImageQualityForHighMag



System Image Quality Confirmation

Spec for Low Mag (23µm & 19µm) :

The vertical edge and horizontal edge value should fall between **3.5 – 10**

Spec for High Mag (11µm) :

The vertical edge and horizontal edge value should fall between **3.5 – 18**

➤ Scroll to latest update date and check on the 3rd column on each of measurement region

Date & Time			Measurement region			Sharpness Result																
192	8/16/2013	6:21:41 PM	top	line	left	86.07	5.685	5.22	top	line	center	86	5.847	5.273	vertical edge	left	2.24	7.138	6.13	horizontal edges	mean	5.246
193	8/16/2013	7:22:07 PM	top	line	left	86.07	5.883	5.403	top	line	center	86.1	5.887	5.369	vertical edge	left	1.77	7.146	6.217	horizontal edges	mean	5.386
194	8/16/2013	8:32:42 PM	top	line	left	86	5.995	5.358	top	line	center	86.17	5.97	5.328	vertical edge	left	2.02	7.162	6.127	horizontal edges	mean	5.343
195	8/16/2013	9:35:16 PM	top	line	left	86.03	5.714	5.215	top	line	center	86	5.795	5.249	vertical edge	left	3.28	6.88	6.132	horizontal edges	mean	5.232
198	8/16/2013	10:41:06 PM	top	line	left	86	5.878	5.322	top	line	center	85.98	5.895	5.374	vertical edge	left	2.4	7.085	5.523	horizontal edges	mean	5.348
199	8/17/2013	1:57:10 PM	top	line	left	86	5.769	5.094	top	line	center	86.17	5.851	5.346	vertical edge	left	1.93	7.298	6.239	horizontal edges	mean	5.22
200			Blank						2			3			4							



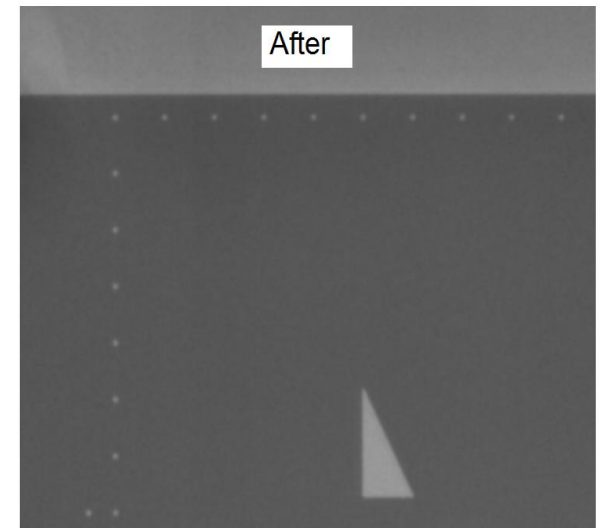
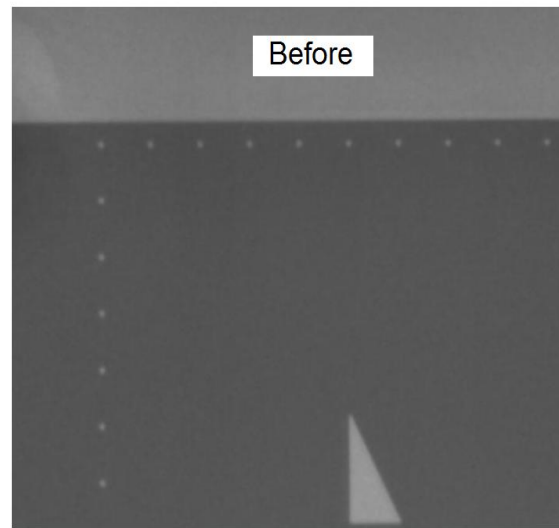
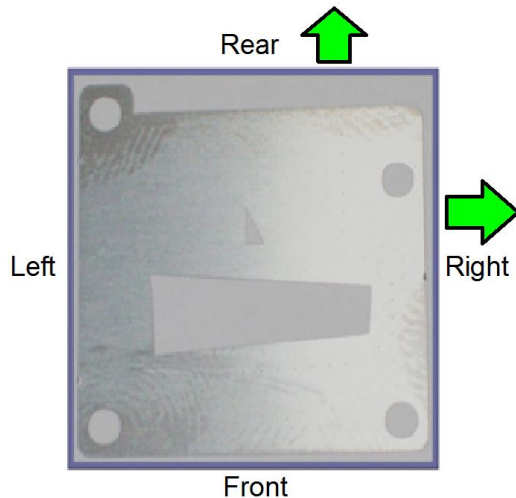
System Image Quality Confirmation

Potential Root Cause:

- X-ray tube position is not optimize
- XY travel stroke not optimize
- X-ray tube itself have problem
- Camera devices have problem
- Fiducial offset cause unable crop correct image for measurement

Solution:

- Re-position the Left Fiducial coupon towards to machine right and rear side





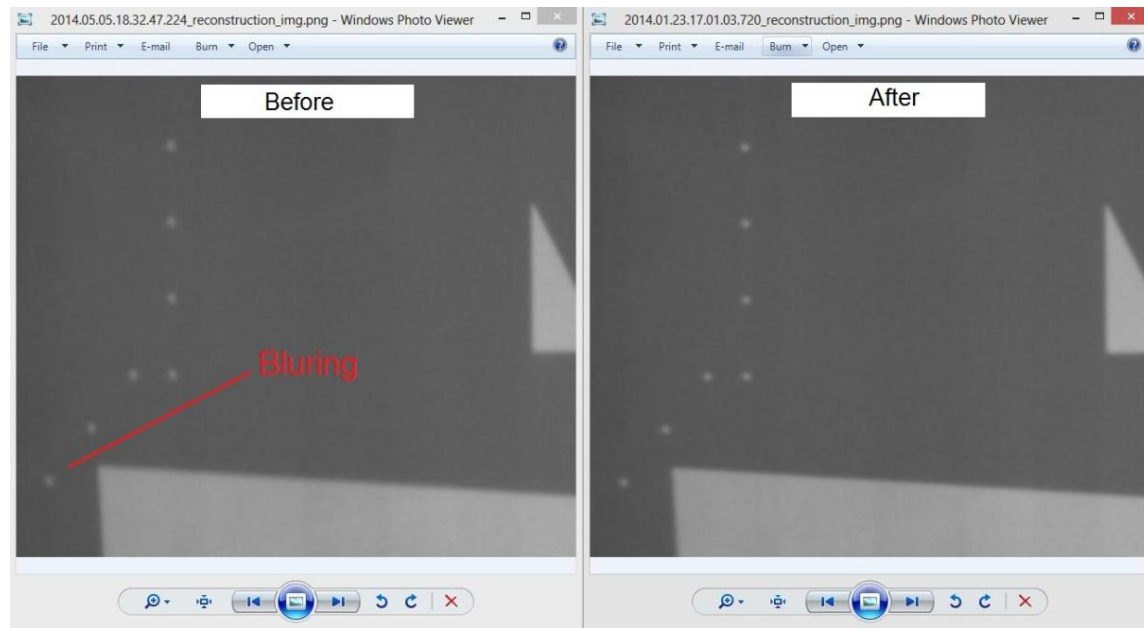
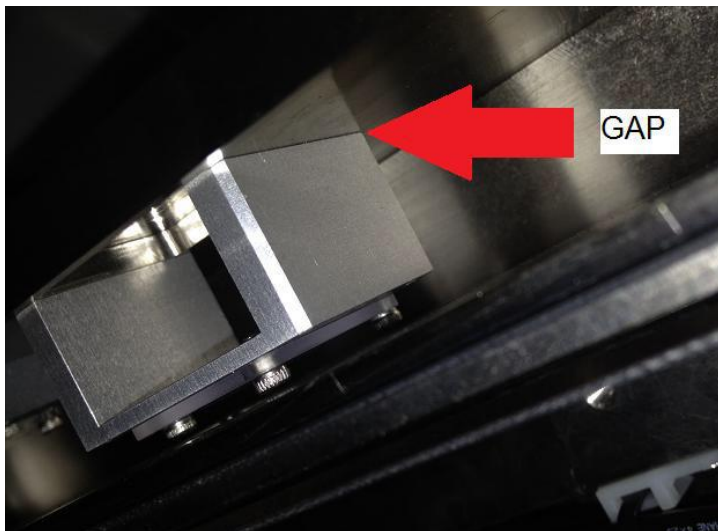
System Image Quality Confirmation Troubleshooting

Potential Root Cause:

Fiducial coupon block slanted/uneven

Solution

Check on Right Fiducial coupon mounting block, ensure no gap. If yes, re-position the mounting block

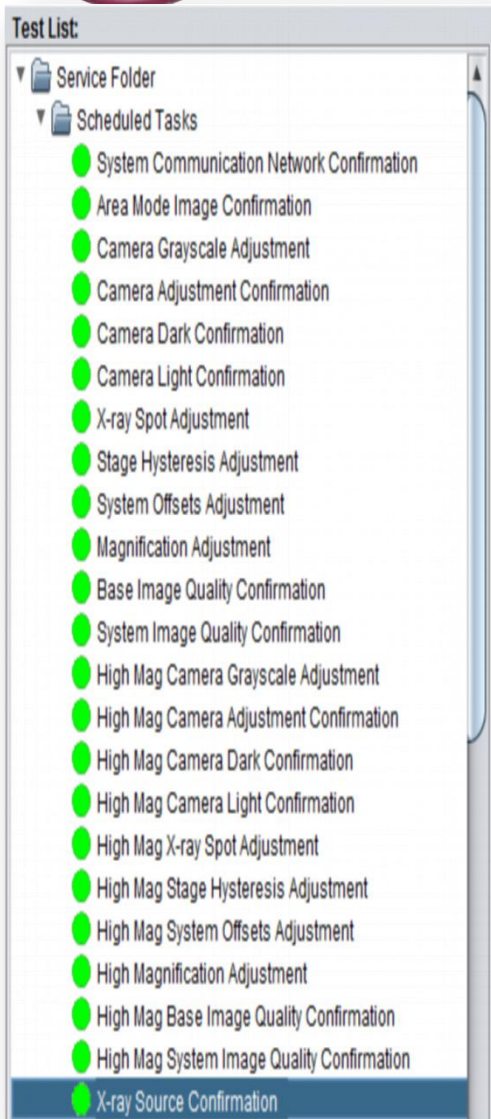




System Image Quality Confirmation Troubleshooting

- **A black or dark image could be caused by the inner barrier not opening, defective x-ray source, improper x-ray spot deflection adjustment**
- **Light gray background with no image could be caused by missing coupon, wrong starting point in the hardware.calib file, a motion of trigger**
- **Vertical line or streaking could be caused by improper greyscale adjustment or debris on the camera array**
- **Horizontal line or streaking could be caused by a motion trigger problem or other image capture issue.**

X-Ray source confirmation



- Purpose** : X-ray source confirmation logs key values of the X-ray tube on a regular basis to allow tracking on these values over time.
- Method** : Runs at start-up and each time the x-rays are turned on or off if its been more than 24 hours
: Logs X-ray On/Off, Anode (kV), Cathode (μ A)
- Results** : Log result to the C:\Program Files\AXI System\v810\5.0\log\confirm\xraySource